

ARTICLE TYPE

Actuarial fire-spreading model based on tree-structured graphical models

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Abstract

In this paper, we propose a general framework to model fire propagation in spatially connected environments, such as adjacent buildings or interconnected rooms within a single dwelling. Each such environment is represented as a tree-like structure, where units (e.g., rooms or substructures) are connected by edges that reflect potential paths for fire spread. Our analysis considers tree-structured sites with up to five units. Since the cases with one, two, and three units are straightforward, we focus on the more intricate configurations with four and five units. For each configuration, we provide a stochastic representation of the total fire losses and analyze the effect of probabilities of propagation using stochastic orders. The investigation compares various tree structures by employing the increasing convex order to highlight differences in risk profiles between them. The model is further examined through a numerical application, offering insights into practical implications for insurance, particularly in calculating premiums and risk measures. Our approach provides a comprehensive tool for assessing fire risk in complex structural layouts, with analytical advantages over simulation-based methods.

Keywords: Graphical models, fire propagation, probability of propagation, fire ignition location.

1. Introduction and literature review

Fire risk modeling is a key component of actuarial science, essential for pricing and risk management. In traditional actuarial approaches, it is sometimes assumed that insured units are exposed to independent fire risks. This assumption becomes questionable in scenarios where fire can propagate between physically connected or adjacent units. It is particularly relevant in densely built environments or within complex structures, where the fire ignition location of one unit may significantly increase the likelihood of damage to others. Modeling such propagation effects is particularly challenging for non-residential buildings constructed with innovative materials, such as cross-laminated timber (CLT). This is due to the limited availability of historical fire loss data. In other words, there are not many such buildings. The lack of data complicates traditional actuarial methods that rely on statistical calibration, as seen in e.g., Wüthrich 2023 and Albrecher, Beirlant, and Teugels 2017.

Given the complexity of modern buildings, which vary in layout, construction materials, and occupancy, fire safety has always been a major concern for protecting life and property. Both civil and fire engineers are interested in the problem of fire hazard modeling, and over time, research has made considerable progress. On one hand, we have early zone models that simplify buildings into homogeneous compartments, which lack precision for predicting localized fire behavior. On the other hand, enhanced tools like the Probabilistic Fire Simulator (PFS) address this limitation by integrating Monte Carlo methods with zone models (see Hostikka and Keski-Rahkonen 2003).

In fire engineering research, models such as Computational Fluid Dynamics (CFD) models, notably the Fire Dynamics Simulator (FDS), have been developed. These models simulate fluid flow, combustion, and heat transfer in detail, see Kallada Janardhan and Hostikka 2019. We have other models like CURisk, which further expanded CFD modeling by incorporating fire growth, smoke movement, evacuation, and economic consequences in timber-frame buildings, see Hadjisophocleous and Fu 2005 and Li, Zhang, and Hadjisophocleous 2013. Lastly, probabilistic models have emerged from Ling's stochastic model of post-flashover spread, see Ling and Williamson 1985, for more advanced Bayesian network approaches for fire risk analysis, see Kim, Dietz, and Matson 2016; Cheng and Hadjisophocleous 2009; Jiang et al. 2025.

While civil and fire engineering emphasize life safety, fire containment, and structural design as outlined in Rychlik and Rydén 2006, actuarial science focuses on loss quantification for pricing and risk management, see Parodi 2023; Wüthrich 2023, which explains the limited number of actuarial models addressing fire propagation. Notable contributions seeking to bridge this gap include Parodi and Watson 2019, who propose a model based on random graphs to describe fire spread, with the aim of linking it to the Bernegger curves introduced in Bernegger 1997, further expanded in Section 4.2 of Parodi et al. 2024. Another significant contribution is the contagion-based model of Boucher et al. 2024, built using insurance data from Canadian farms to capture spatial and temporal dependencies in fire dynamics. These models represent early but important steps toward integrating structural, spatial, and mechanistic considerations into actuarial fire loss modeling.

The role of building design, material choice, and suppression systems remains central in fire behavior. Open layouts and combustible materials promote fire spread, whereas fire barriers and sprinklers limit it, see Hurley et al. 2015; Cowlard et al. 2013. In addition, recent findings on CLT walls have revealed uncertainties in fire resistance, highlighting the need for adapted modeling strategies in such contexts, see Wiesner et al. 2022.

In this study, we propose an actuarial fire-spreading model to understand better fire behavior in interconnected residential and commercial building structures. Our model is based on the physical intuition of fire propagation between adjacent units, using an underlying tree structure to represent dependencies. This aligns with the philosophy advanced by Parodi et al. 2024, who argue that loss models should, whenever possible, be rooted in the actual mechanisms of loss generation rather than chosen solely for statistical fit.

The paper is structured as follows. We start with the desired properties and ingredients of fire-spreading models in Section 2, then we provide some general notions on probabilistic graphical models in Section 3, then we move to the Actuarial fire-spreading model itself in Section 4. In Section 5 we introduce the Actuarial fire-spreading model and for a building with various number of unit and then we build the model for a building with 5 units. Finally, in Section 6, we introduce the Maxwell Boltzmann Bose Einstein Fermi Dirac distribution (MBBEFD) and then make some fits of the MBBEFD distribution to the analytically derived distribution of losses from our model.

2. Desired properties and ingredients of fire-spreading models

While our general modeling framework is applicable to a wide range of settings, such as adjacent buildings in dense urban environments or interconnected industrial units, we now restrict our focus to a more specific case. So, in the remainder of this paper, we focus on fire propagation between rooms or compartments located on the same floor of a single building.

In actuarial modeling of fire risk, a key quantity of interest is the aggregate loss amount X_ξ due to fire hazards in a given building with exposure ξ , typically measured in square meters of floor area. This loss can be represented as

$$X_\xi = \begin{cases} \sum_{k=1}^{M_\xi} B_k, & M_\xi > 0, \\ 0, & M_\xi = 0, \end{cases}$$

where $M_\xi \sim \text{Pois}(\xi\lambda)$ denotes the number of fire incidents in the building, and $\{B_k, k = 1, 2, \dots\}$ is a sequence of independent and identically distributed (i.i.d.) random variables (rvs) representing the loss amount from each fire. These rvs are assumed independent of M_ξ , and identically distributed with cdf F_B .

The parameter λ is known as the *fire incident rate*. It represents the average number of fires per unit of floor area per year (e.g., per square meter per year). It quantifies the frequency of fire occurrence in buildings of a given type. Typically, it serves as an input for risk assessment models. Gaskin and Yung 1993 provide for instance the fire incident rate for office buildings in the U.S.A. That was estimated at $\lambda = 7.30 \times 10^{-6}$ fires per square meter per year. That means a supermarket with a floor area of $\xi = 1500 \text{ m}^2$ would be expected to experience $\xi\lambda = 0.01095$ fires per year, on average.

2.1 Desired properties and key ingredients of fire-spreading models

To be useful in modern actuarial practice, fire-spreading models must meet both conceptual and physical requirements. First of all, they should conceptually be adaptable to advances in civil and fire engineering, then they should be flexible enough to model diverse building types and new materials like CLT, and lastly, they should remain interpretable and implementable for practical insurance use.

Physically, such models must incorporate critical factors influencing fire behavior, including fuel load, compartment geometry, ventilation, ignition source and location, and the type of fire as outlined in Cheng and Hadjisophocleous 2009. By combining these methodological and physical elements, fire-spreading models can provide robust, scientifically grounded tools for actuarial fire risk analysis.

2.2 Our modeling perspective

Our model captures the dynamics of fire propagation in complex building configurations, drawing from actuarial reasoning and fire safety engineering. In Cheng and Hadjisophocleous 2009 for example the authors employ Bayesian networks to represent fire propagation from a known fire ignition location through adjacent compartments. In such models, fire spreads are governed by the structure of a directed acyclic graph (DAG), representing conditional dependencies.

Our approach builds on this idea but reinterprets fire spreading through the lens of *Markov random fields* (MRFs). This shift offers our model greater flexibility for representing symmetrical dependencies and allows for a natural or realistic interpretation of spatial fire propagation in building networks. As explained in Chapter 4 of Koller and Friedman 2009, there exists a close connection between Bayesian networks and MRFs. This allows insights and techniques from one framework to inform the other. By adopting this approach, we aim to propose a fire-spreading model that is both theoretically sound and practically applicable in insurance contexts where historical data is sparse or unavailable.

The general objective of this paper is to study fire propagation within such structures using graphical models in order to model the resulting fire losses. To achieve this, we pursue three specific objectives: first, we introduce a fire-spreading model based on tree-structured graphical models and analyze its theoretical properties; second, we study the distribution of fire losses generated by this model; and lastly, we calibrate its components by accounting for structural characteristics (e.g., wall layout, door positioning), passive fire protection systems, and explore practical applications. By explicitly incorporating fire propagation mechanisms, our model bridges a crucial gap between engineering insights and actuarial practice, combining concepts from civil engineering and actuarial science to provide a more comprehensive tool for assessing fire risk in modern construction.

3. Probabilistic graphical models

Probabilistic graphical models (PGMs) provide a flexible framework for representing dependencies among multivariate rvs by associating variables to vertices and encoding dependencies through

edges, see Lauritzen 1996, Wainwright, Jordan, et al. 2008, and Koller and Friedman 2009 for comprehensive treatments. PGMs fall into two main categories: *Bayesian networks* (directed acyclic graphs), which have been used to model fire propagation from an ignition point in Cheng and Hadjisophocleous 2009. We also have *MRFs*, which are based on undirected graphs and suited for reciprocal interactions.

We adopt an approach based on tree-structured (MRFs) with Bernoulli marginals to model fire spread. In this setting, each vertex represents a unit within a building (e.g., room). Each vertex is also associated with a binary variable indicating whether it is affected by fire, and edges capture the potential for fire to propagate. The graph structure allows for flexible modeling of spatial dependence between units, and the tree assumption ensures analytical tractability. Moreover, using Bernoulli marginals naturally accommodates binary fire outcomes (burned vs. not burned).

Since we rely on graph theory as the principal mathematical tool for our fire-spreading model, we review the essential graph-theoretic definitions and notation that will be used throughout the paper before building the model as an MRF.

- A **graph** $\mathcal{G}$ is a pair $(\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \dots, d\}$ is a set of d vertices and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of edges linking the vertices. An *undirected graph* means that for any $u, v \in \mathcal{V}$, if $(u, v) \in \mathcal{E}$, then $(v, u) \in \mathcal{E}$ as well.
- A **path** from vertex u to vertex v , denoted $\text{path}(u, v)$, is a sequence of successive edges in $\mathcal{E}$, starting at u and ending at v , where each edge appears at most once. A vertex lies on the path if it is part of an edge in that sequence. The set of vertices on the path from u to v is denoted by $\mathcal{V}_{\text{path}(u,v)}$.
- A **tree**, denoted $\mathcal{T} = (\mathcal{V}, \mathcal{E})$, is a connected, undirected, acyclic graph. It contains exactly $d - 1$ edges and connects every pair of vertices by a unique path. A tree can be rooted at any vertex $r \in \mathcal{V}$, yielding a rooted tree $\mathcal{T}_r$, where r is the *root*. Each vertex $v \neq r$ has a unique *parent* $\text{pa}(v)$, defined as the only adjacent vertex that is not a descendant of v .
- The **descendants** of v , denoted $\text{desc}(v)$, are the vertices whose path to the root passes through v . Among them, the *children* $\text{ch}(v)$ are the ones directly connected to v .
- A **leaf** is a vertex with no children, and the root has no parent.

To illustrate these notions, Figure 1a shows an undirected graph that is not a tree, as it contains cycles. In contrast, Figure 1b presents a rooted tree structure, which will be central to our modeling framework. In the tree from Figure 1b, we have $\mathcal{V} = \{A, B, C, D, E, F, G, H, I\}$ and $\mathcal{E} = \{(A, B), (A, C), (B, D), (B, E), (C, F), (C, G), (C, H), (F, I)\}$. The root is A , and the leaves are D, E, G, H , and I . Also,

- $\text{desc}(C) = \{F, G, H, I\}$ and $\text{ch}(C) = \{F, G, H\}$;
- $\text{pa}(F) = C$ and $\text{pa}(I) = F$;
- $\text{path}(A, I) = \{(A, C), (C, F), (F, I)\}$ with $\mathcal{V}_{\text{path}(A,I)} = \{A, C, F, I\}$.

Trees and graphs are indispensable tools for modeling and analyzing complex systems. For more details on their properties, see Chapter 4 of Balakrishnan and Ranganathan 2012.

Modeling fire propagation as a probabilistic process on a tree structure reflects the network-like organization of units within the site. At the same time, probabilities of propagation characterize the stochastic nature of fire propagation between connected units. In addition, this approach aligns with terminology commonly used in the literature. Terms such as *fire propagation* see Parodi and Watson 2019, *fire spreading* see Cheng and Hadjisophocleous 2009, and *contagion* see Boucher et al. 2024 are often used interchangeably to describe the same process of fire transmission between connected units. Throughout this paper, we will use the term *fire propagation* to refer to the process of fire transmission between units. We recall the characterization of MRFs on graphs based on the *global Markov property*.



Figure 1. Example of graph and tree

Definition 1. Let $X = \{X_v, v \in \mathcal{V}\}$ be a vector of random variables defined on a tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$. We say that X is an MRF if it satisfies the local Markov property: for any two components X_u and X_v of X such that $(u, v) \notin \mathcal{E}$, we have $X_u \perp\!\!\!\perp X_v \mid \{X_w : w \in S(u, v)\}$, where $S(u, v)$ is a separator for the non-adjacent vertices u and v , see Chapter 3 of Lauritzen 1996. Where $\perp\!\!\!\perp$ denotes the conditional independence. More precisely, $S(u, v)$ is a subset of $\mathcal{V}$ whose removal from $\mathcal{T}$ results in u and v belonging to two distinct subtrees. On a tree $\mathcal{T}$, any single vertex on path (u, v) can act as a separator to $\{u\}$ and $\{v\}$.

As mentioned in Lauritzen 1996, the global Markov property implies the local Markov property while the latter implies the pairwise Markov property.

4. Actuarial fire-spreading model

4.1 Definitions

In this section, we propose our actuarial fire-spreading model (AFSM). Consider the floor of a site, which may represent a building (residential or commercial), a farm, or a construction site. The floor is modeled as a structure composed of d distinct units organized in a tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \dots, d\}$ is the set of vertices representing the units, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges indicating the potential connections between these units. In this context, each vertex $v \in \mathcal{V}$ corresponds to a specific unit v within the structure, where the term *unit* refers to an individual room or compartment, such as one section of a building or part of a construction site. Since we only consider trees, there is no path from a vertex to itself.

We represent in Figure 2 the floor layout using a tree, following the visual modeling approach introduced in Li, Zhang, and Hadjisophocleous 2013 to capture the spatial configuration and potential fire propagation paths.

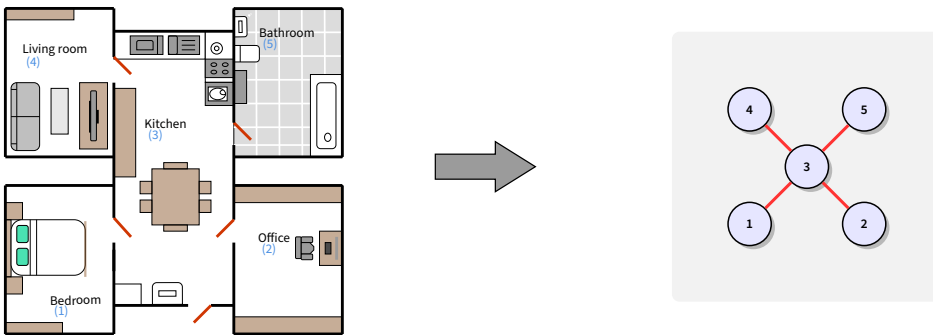


Figure 2. Site layout and corresponding tree representation in hub

Since the fire propagation dynamics between units are governed by the tree structure, it restricts

the propagation of fire to adjacent units based on the edges in $\mathcal{E}$. Specifically, a fire can only propagate from unit v_1 to unit v_2 or similarly from unit v_2 to unit v_1 if there exists an edge $(v_1, v_2) \in \mathcal{E}$. This indicates that these two units are directly connected. Between edges, we use the symbol $\rightarrow$ to indicate the direction of the fire propagation. Each edge $(v_1, v_2) \in \mathcal{E}$ is associated with a probability of propagation denoted $p_{v_1 \rightarrow v_2}$ (or equivalently $p_{v_2 \rightarrow v_1}$) that quantifies the probability that the fire propagates from unit v_1 to the connected unit v_2 . The probability that the fire does not propagate from the unit v_1 to the connected unit v_2 equals $1 - p_{v_1 \rightarrow v_2}$. Equivalently, the probability that the fire does not propagate from unit v_2 to the connected unit v_1 equals $1 - p_{v_2 \rightarrow v_1}$. This probability depends on various factors, such as the physical proximity of the units, the nature of the materials separating them, and the effectiveness of fire barriers between units. For a detailed overview about the computation of those probabilities, we refer to Cheng 2011. A synthesis of the latter can be found in Cheng and Hadjisophocleous 2009. See also Ling and Williamson 1985 for an initial study on this topic and Jiang et al. 2025 for a recent one.

We now introduce the adjacency matrix associated with a given structure $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ of d units. Let $\mathbf{A}$ be a $d \times d$ matrix whose element at row v_1 and column v_2 is defined by

$$a(v_1, v_2) = \begin{cases} p_{v_1 \rightarrow v_2}, & \text{if } (v_1, v_2) \in \mathcal{E}, \\ 0, & \text{otherwise.} \end{cases}$$

Since we consider undirected trees, each edge $(v_1, v_2) \in \mathcal{E}$ represents a symmetric propagation relationship, meaning that $p_{v_1 \rightarrow v_2} = p_{v_2 \rightarrow v_1}$. In this case, $\mathbf{A}$ is symmetric. To avoid redundancy in notation, we may assume without loss of generality that $v_1 < v_2$ when indexing edges. To illustrate, consider the case $d = 4$ for which there are only two distinct tree structures possible: the series configuration $\mathcal{T}^{4,1}$ and the hub configuration $\mathcal{T}^{4,2}$, both shown in Figure 10. Their corresponding adjacency matrices are the following:

$$\mathbf{A}^{\mathcal{T}^{4,1}} = \begin{pmatrix} 0 & p_{1 \rightarrow 2} & 0 & 0 \\ p_{2 \rightarrow 1} & 0 & p_{2 \rightarrow 3} & 0 \\ 0 & p_{3 \rightarrow 2} & 0 & p_{3 \rightarrow 4} \\ 0 & 0 & p_{4 \rightarrow 3} & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{A}^{\mathcal{T}^{4,2}} = \begin{pmatrix} 0 & p_{1 \rightarrow 2} & 0 & 0 \\ p_{2 \rightarrow 1} & 0 & p_{2 \rightarrow 3} & p_{2 \rightarrow 4} \\ 0 & p_{3 \rightarrow 2} & 0 & 0 \\ 0 & p_{4 \rightarrow 2} & 0 & 0 \end{pmatrix}.$$

Additionally, to each unit $v \in \mathcal{V}$, let us associate a Bernoulli (or binary) rv I_v , which takes the value 1 if unit v is affected by fire, and 0 otherwise.

Let $\mathbf{I} = (I_v, v \in \mathcal{V})$ be a d -dimensional vector of fire indicators with joint probability mass function (pmf) of $\mathbf{I}$ is denoted by $f_{\mathbf{I}}$.

The specification of this joint distribution will be developed in the next section using our AFSM, which is inspired by several fire propagation models found in the civil and fire engineering literature.

4.2 Fire ignition location

Fires can originate from one or more locations within a building. In fire safety literature, sources of fire are typically associated with ignition hazards, such as open flames, electrical faults, or overheated appliances, see Buchanan and Abu 2017; Kodur, Kumar, and Rafi 2020. These ignition sources can be favored by precursor conditions like flammable liquids or faulty wiring. Once a fire starts, its spread is influenced by propagation hazards such as the presence of combustible materials, whose thermal properties determine how quickly fire can grow, see Guerrazzi 2022. According to Kodur, Kumar, and Rafi 2020, cooking remains one of the most common causes of ignition in both residential and non-residential buildings.

Let Θ denote a discrete rv representing the fire ignition location. In the AFSM, for a known fire ignition location, we assume that a room being affected by fire results from that single, unique origin.

It means that, if a fire occurs in structure $\mathcal{T}$, it cannot be the result of two or more fire ignition locations (units). The pmf of discrete rv Θ is denoted by f_{Θ} , where $f_{\Theta}(s) = \Pr(\Theta = s) \geq 0$, for $s \in \mathcal{V}$. The conditional joint pmf of $\mathbf{I}$ given that the fire originates in room s , is denoted by

$$f_{\mathbf{I}|\Theta=s}(\mathbf{i}) = \Pr(I_1 = i_1, \dots, I_d = i_d | \Theta = s), \quad \text{for } \mathbf{i} = (i_1, \dots, i_d) \in \{0, 1\}^d.$$

Assume fire ignites in unit s . It may flashover in that unit with probability $p_s \in (0, 1)$ defined as

$$\Pr(I_s = 1 | \Theta = s) = p_s,$$

or die out at the source leaving the unit s intact, with probability q_s where

$$\Pr(I_s = 0 | \Theta = s) = q_s = 1 - p_s.$$

If the fire successfully flashes over in unit s , it can then spread to adjacent units and continue spreading along the connections defined by the edge set $\mathcal{E}$ of the structure $\mathcal{T}$.

In our model, if we consider the fire ignition location $\Theta = s$, the vector $\mathbf{I}$ (indicating which units are burning) given Θ is an MRF which means that the probability of fire propagation from a given unit depends only on its current state (burned or not) and not on the trajectory the fire took to reach it: fire propagation is local, and each unit only depends on its neighbors. However, conditioning on Θ , the joint pmf of $\mathbf{I}$ is

$$f_{\mathbf{I}}(\mathbf{i}) = \sum_{s \in \mathcal{V}} f_{\Theta}(s) \times f_{\mathbf{I}|\Theta=s}(\mathbf{i}), \quad \text{for } \mathbf{i} \in \{0, 1\}^d. \quad (1)$$

The pmf given in (1) corresponds to the pmf of a mixed distribution. But mixtures do not, in general, preserve conditional independencies: different ignition points induce different dependency patterns, and averaging over them introduces additional long-range dependencies. As a result, the unconditional distribution of $\mathbf{I}$ is not itself an MRF, since the Gibbs factorization

$$f_{\mathbf{I}}(\mathbf{i}) = \prod_{v \in \mathcal{V}} f_{I_v}(i_v) \times \prod_{(v_1, v_2) \in \mathcal{E}} \frac{f_{I_{v_1}, I_{v_2}}(i_{v_1}, i_{v_2})}{f_{I_{v_1}}(i_{v_1}) f_{I_{v_2}}(i_{v_2})}$$

does not hold for $f_{\mathbf{I}}$ as in Equation 1.

4.3 Construction of the model

To describe the behavior of fire propagation within a structure, we derive in this section the conditional joint probability mass function of a random vector indicating the fire status at each vertex. We consider separately the cases where the origin of the fire, which we designate by unit $s \in \mathcal{V}$, is at root of the tree or at another node of the tree.

4.3.1 Fire propagation dynamics from a root in a tree structure

Let us assume node s to be the root of the tree $\mathcal{T}$, resulting in the rooted tree $\mathcal{T}_s$. This rooted structure allows for a natural representation of the directionality of fire spread, from the fire ignition location toward the other units. This means that the potential propagation from a unit u to another unit v is only considered if u is the parent of v in $\mathcal{T}_s$, denoted $\text{pa}(v) = u$. The fire spread hence possesses somewhat of a hierarchical structure in the rooted tree, simplifying the specification of the conditional propagation probabilities. For each descendant v of s , the probability that v is affected by fire, given the state of its parent and the fire ignition location at s , will be precisely defined in this section.

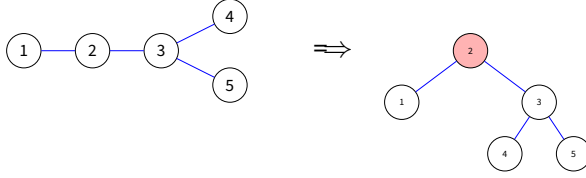


Figure 3. Binary tree with 5 nodes, with ignition at Node 2, tree rooted 2

Let $\mathbf{I} = (I_\nu)_{\nu \in \mathcal{V}}$ denote the vector of binary random variables indicating the fire status of each vertex in the tree structure. Conditional on the fire ignition location $s \in \mathcal{V}$, the joint pmf of $\mathbf{I}$ can be expressed as

$$f_{\mathbf{I}|\Theta=s}(\mathbf{i}) = \Pr(I_s = i_s | \Theta = s) \prod_{\nu \in \text{desc}(s)} \Pr(I_\nu = i_\nu | I_{\text{pa}(\nu)} = i_{\text{pa}(\nu)}), \quad \text{for } \mathbf{i} \in \{0, 1\}^d, \quad (2)$$

which shows that $(\mathbf{I}|\Theta = s)$ follows a Gibbs distribution (see Chapter 4 of Koller and Friedman 2009 for a detailed discussion). Similarly, for any subset $\mathcal{W} \subset \mathcal{V}$, the conditional pmf of $\mathbf{I}_{\mathcal{W}} = \{I_\nu, \nu \in \mathcal{W}\}$ can be obtained from (2).

The decomposition in (2) relies on the local propagation mechanism of the fire through the tree considering the fact that fire ignition location is the root of the tree, colored in pink in Figure 3. The fire hence starts at node 2. Each vertex ν depends only on the state of its parent $\text{pa}(\nu)$ and the propagation probability along the edge $\text{pa}(\nu) \rightarrow \nu$. More precisely, for all $\nu \in \text{desc}(s)$, we have

$$\Pr(I_\nu = i_\nu | I_{\text{pa}(\nu)} = i_{\text{pa}(\nu)}) = \begin{cases} 1, & (i_{\text{pa}(\nu)}, i_\nu) = (0, 0), \\ 1 - p_{\text{pa}(\nu) \rightarrow \nu}, & (i_{\text{pa}(\nu)}, i_\nu) = (1, 0), \\ 0, & (i_{\text{pa}(\nu)}, i_\nu) = (0, 1), \\ p_{\text{pa}(\nu) \rightarrow \nu}, & (i_{\text{pa}(\nu)}, i_\nu) = (1, 1). \end{cases} \quad (3)$$

Equation (3) simply encodes the propagation dynamics: if the parent vertex is not on fire ($I_{\text{pa}(\nu)} = 0$), then the fire cannot reach ν ($I_\nu = 0$ with probability 1). Conversely, if the parent is on fire ($I_{\text{pa}(\nu)} = 1$), the fire spreads to ν with probability $p_{\text{pa}(\nu) \rightarrow \nu}$, and fails to spread with probability $1 - p_{\text{pa}(\nu) \rightarrow \nu}$. The model therefore prohibits spontaneous ignitions, enforcing that $I_\nu = 1$ only if $I_{\text{pa}(\nu)} = 1$.

4.3.2 Propagation dynamics when the fire starts at a non-root vertex

Let us now consider the case where the fire starts at a vertex $s \in \mathcal{V}$ that is *not* the root. As illustrated in a tree rooted at 2 represented in Figure 4a, the fire ignition location is the node 5 and $\mathcal{V}_{\text{path}(2,5)} = \{2, 3, 5\}$ corresponds to the circled branch of tree. To study this scenario, we can consider that the fire ignition location $s \neq r$ is the root of the subtree starting there. This re-rooting allows us to express the propagation probabilities in a form analogous to the root-ignition case. This is the case for the subtree rooted 3 in Figure 4b.



Figure 4. Rooted binary tree ($r = 2$), with 5 nodes and different ignition locations

With the same reasoning as in Section 4.3.1, we have

$$f_{I|\Theta=s}(i) = \Pr(I_r = i_r | \Theta = s) \prod_{v \in \text{desc}(r)} \Pr(I_v = i_v | I_{\text{pa}(v)} = i_{\text{pa}(v)}, \Theta = s), \quad \text{for } i \in \{0, 1\}^d. \quad (4)$$

The expression of $\Pr(I_v = i_v | I_{\text{pa}(v)} = i_{\text{pa}(v)}, \Theta = s)$ in (4) depends on the location of v in the rooted tree $\mathcal{T}_r$, whether it is outside or inside the path between the fire ignition location s and the root r . For that reason, we need to split the $\text{desc}(r)$ in two mutually exclusive subsets. Firstly, we define the subset $A_{s,r} = \mathcal{V}_{\text{path}(s,r)} \setminus \{r\}$ of all the nodes that belong to the path between the fire ignition location node s and the root r , excluding the root itself. Secondly, $B_{s,r} = \text{desc}(r) \setminus A_{s,r}$ is the subset of all nodes among the descendants of the root r that are not in the subset $A_{s,r}$. Hence, $A_{s,r} \cup B_{s,r} = \text{desc}(r)$ and $A_{s,r} \cap B_{s,r} = \emptyset$. It follows that

$$\Pr(I_r = 1 | \Theta = s) = \Pr(I_s = 1 | \Theta = s) \times \prod_{(u,w) \in \text{path}(s,r)} p_{uw}$$

and

$$\Pr(I_r = 0 | \Theta = s) = 1 - \Pr(I_r = 1 | \Theta = s).$$

Then, for all $v \in A_{s,r}$, we have

$$\Pr(I_v = i_v | I_{\text{pa}(v)} = i_{\text{pa}(v)}, \Theta = s) = \begin{cases} \frac{\Pr(I_{\text{pa}(v)}=0, I_v=0 | \Theta=s)}{\Pr(I_{\text{pa}(v)}=0 | \Theta=s)}, & (i_{\text{pa}(v)}, i_v) = (0, 0), \\ 0, & (i_{\text{pa}(v)}, i_v) = (1, 0), \\ \frac{\Pr(I_{\text{pa}(v)}=0, I_v=1 | \Theta=s)}{\Pr(I_{\text{pa}(v)}=0 | \Theta=s)}, & (i_{\text{pa}(v)}, i_v) = (0, 1), \\ 1, & (i_{\text{pa}(v)}, i_v) = (1, 1), \end{cases} \quad (5)$$

where $\Pr(I_{\text{pa}(v)} = 0 | \Theta = s) = \Pr(I_v = 0, I_{\text{pa}(v)} = 0 | \Theta = s) + \Pr(I_v = 1, I_{\text{pa}(v)} = 0 | \Theta = s)$, assuming that $\Pr(I_v = 0, I_{\text{pa}(v)} = 0 | \Theta = s) > 0$ and $\Pr(I_v = 1, I_{\text{pa}(v)} = 0 | \Theta = s) > 0$.

Finally, for all nodes $v \in B_{s,r}$, we find

$$\Pr(I_v = i_v | I_{\text{pa}(v)} = i_{\text{pa}(v)}, \Theta = s) = \begin{cases} 1, & (i_{\text{pa}(v)}, i_v) = (0, 0), \\ 1 - p_{\text{pa}(v),v}, & (i_{\text{pa}(v)}, i_v) = (1, 0), \\ 0, & (i_{\text{pa}(v)}, i_v) = (0, 1), \\ p_{\text{pa}(v),v}, & (i_{\text{pa}(v)}, i_v) = (1, 1). \end{cases} \quad (6)$$

In this framework, Equation (6) gives the propagation dynamic of the fire for a vertex v in $\mathcal{T}_r$. This is done by conditioning on the state of its parent $\text{pa}(v)$, when the fire originates at s and v is a descendant of the chosen root r but does not lie on the direct path from r to s ($v \in B_{s,r}$).

We first start with the case where $I_{pa(v)} = 0$ and $I_v = 0$, which means neither the parent nor v itself is burning. We can note that this is always possible since no fire can reach v if its parent is unaffected. This occurs with probability 1. We then proceed with the case where the parent vertex is burning, but the child vertex is not ($I_{pa(v)} = 1$ but $I_v = 0$), which occurs if the fire fails to spread from the parent to the child. It happens with probability $1 - p_{pa(v) \rightarrow v}$. Lastly, we look at the situation where both the parent and the child are burning, which occurs when $I_{pa(v)} = 1$ and $I_v = 1$. It is possible when the fire successfully propagates from the parent to the child, and it happens with probability $p_{pa(v) \rightarrow v}$. Evidently, if $I_{pa(v)} = 0$ and $I_v = 1$, this case is impossible because a child cannot burn if its parent is unaffected, resulting in a probability of 0.

In Equation (5), we describe the fire propagation dynamic for a vertex v that lies on the direct path from the root r to the fire ignition location s excluding the root itself, the set we denoted $A_{s,r}$. In Figure 4, the circled branches of the trees represent the set of nodes lying on the path between the fire ignition locations and the root.

First of all, we consider the case where neither the parent nor the vertex itself is burning ($I_{pa(v)} = 0$ and $I_v = 0$). This means in this case that fire died before reaching v . The probability that the fire also does not reach the child node v is given by the proportion of cases where both nodes remain unaffected. In other words: among all scenarios where the parent is safe, this fraction counts how many of those have the child also safe.

Next, if $I_{pa(v)} = 1$ but $I_v = 0$, this indicates that the fire reached the parent but not the child, which is not possible along this critical path, where the fire must propagate continuously to reach $pa(v)$. In Figure 4a, fire cannot reach node 2 without passing through node 3 for example. Thus, this case has a probability of 0.

If $I_{pa(v)} = 0$ but $I_v = 1$, meaning the node is on fire while the parent is not. Since the fire ignition location $\Theta = s$ lies in the subtree for which v is an ancestor, it is possible for the fire to reach node v without reaching its parent $pa(v)$. In this situation, the fire may propagate upward from s to v , and then extinguish along the edge connecting v to $pa(v)$. In this context, the ratio in 5 represents the proportion, among all scenarios in which the parent node is not reached, of those in which the fire nevertheless reaches v . The numerator counts precisely the situations in which the fire climbs up to v and dies out before reaching $pa(v)$, whereas the denominator includes all scenarios in which the propagation stops before the parent, regardless of whether v was reached or not.

Finally, if $I_{pa(v)} = 1$ and $I_v = 1$, this means that both the parent and the vertex v are burning. Since we have a tree structure and a unique path from s to any vertex, if the parent is on fire and we know that the fire originates from vertex s , this means the fire must have passed through vertex v to reach its parent. For example in Figure 4b, node 2 is on fire only if fire propagates from node 5 to node 3 and from node 3 to node 2. This configuration occurs with probability 1.

4.3.3 Equivalence of conditional pmf expressions

We now show that the two conditional pmfs given in (2) and (4) are in fact equivalent. In particular, choosing $r = s$ yields Equation (2), while choosing any other vertex $r \in \mathcal{V}$ gives Equation (4).

Let $\text{path}(s, r) = (v_0 = s, v_1, \dots, v_k = r)$ be the unique path from s to r , with v_j denoting the j -th vertex along this path. By the global Markov property on the tree, we can write:

$$\Pr(I_r = i_r \mid \Theta = s) = \Pr(I_{v_k} = i_{v_k} \mid I_{v_{k-1}} = i_{v_{k-1}}, \Theta = s) \cdots \Pr(I_{v_1} = i_{v_1} \mid I_{v_0} = i_{v_0}, \Theta = s) \Pr(I_{v_0} = i_{v_0} \mid \Theta = s).$$

In compact form, this becomes

$$\Pr(I_r = i_r \mid \Theta = s) = \Pr(I_s = i_s \mid \Theta = s) \prod_{v \in \text{path}(s, r) \setminus \{s\}} \Pr(I_v = i_v \mid I_{pa(v)} = i_{pa(v)}, \Theta = s). \quad (7)$$

Inserting (7) into Equation (4), we obtain

$$f_{I|\Theta=s}(i) = \left(\Pr(I_s = i_s \mid \Theta = s) \prod_{\nu \in \text{path}(s,r) \setminus \{s\}} \Pr(I_\nu = i_\nu \mid I_{\text{pa}(\nu)} = i_{\text{pa}(\nu)}, \Theta = s) \right) \\ \times \prod_{\nu \in \text{desc}(r)} \Pr(I_\nu = i_\nu \mid I_{\text{pa}(\nu)} = i_{\text{pa}(\nu)}, \Theta = s).$$

The union of vertices in $\text{path}(s,r) \setminus \{s\}$ and $\text{desc}(r)$ is exactly $\text{desc}(s)$. Since fire spreads from s to r and then from r to its descendants.

Hence, we recover exactly the factorization in Equation (2):

$$f_{I|\Theta=s}(i) = \Pr(I_s = i_s \mid \Theta = s) \prod_{\nu \in \text{desc}(s)} \Pr(I_\nu = i_\nu \mid I_{\text{pa}(\nu)} = i_{\text{pa}(\nu)}, \Theta = s), \quad \text{for } i \in \{0, 1\}^d.$$

4.4 Severity variables and independence assumption

We define the d -dimensional vector of independent rvs $\mathbf{Y} = (Y_\nu, \nu \in \mathcal{V})$. $Y_\nu \in [0, c_\nu]$ represents the amount of damage in unit ν , $\nu \in \mathcal{V}$, and c_ν is the property value (or insured value) of unit ν . The rv Y_ν can be either discrete or continuous, with cdf $F_{Y_\nu}(\gamma) = \Pr(Y_\nu \leq \gamma)$, $0 \leq \gamma \leq c_\nu$, and $F_{Y_\nu}(\gamma) = 1$, $\gamma \geq c_\nu$. We assume that the vectors of rvs $\mathbf{I}$ and $\mathbf{Y}$ are independent, indicating that the fire occurrence and the resulting losses due to the fire are independent events.

Finally, we define the vector of d rvs $\mathbf{D} = (D_\nu, \nu \in \mathcal{V})$, where

$$D_\nu = \begin{cases} Y_\nu, & I_\nu = 1, \\ 0, & I_\nu = 0. \end{cases}$$

The independence assumption between I_ν and Y_ν , $\nu \in \mathcal{V}$, leads to the following expected value of D :

$$E[D_\nu] = E[Y_\nu] \times E[I_\nu],$$

which corresponds to the expected amount of damage in unit ν if a fire starts or propagates to unit ν , $\nu \in \mathcal{V}$.

The total losses for the entire site caused by a single fire event are represented by the positive rv B , defined as

$$B = \sum_{\nu \in \mathcal{V}} D_\nu = \sum_{\nu \in \mathcal{V}} I_\nu \times Y_\nu. \quad (8)$$

If $(B|\Theta = s)$ is the total losses of the entire site caused by a fire starting at unit $\Theta = s$, then (8) admits the representation

$$B = \sum_{s \in \mathcal{V}} \mathbb{1}_{\{\Theta=s\}} (B|\Theta = s). \quad (9)$$

The cdf of B obtained by conditioning on the unit in which the fire was ignited, that is

$$F_B(x) = \sum_{s \in \mathcal{V}} f_\Theta(s) F_{B|\Theta=s}(x), \quad x \in [0, b], \quad (10)$$

given (8) and $F_B(x) = 1$ for $x > b$, where $b = \sum_{\nu \in \mathcal{V}} c_\nu$ is the total property value (or insured value) of the site.

In the next section, we provide a detailed description of the AFSM within a site of $d = 5$. By focusing on this case, we can explore the intricacies of fire propagation within that specific structural configurations

5. Actuarial fire spread model for sites with various numbers of units

To ground the model in a more concrete setting, we derive the necessary equations for tree-structured sites with various values of d . In [Appendix 1](#), adapting the configurations shown in [Appendix 3](#) of [Harary 2018](#), we list all trees for $d = 4$ and $d = 5$. The cases with $d < 4$, ($d = 1, 2, 3$) are relatively easy to deal with: for $d = 1$, fire cannot propagate due to a single, isolated unit; for $d = 2$ and $d = 3$, the structure reduces to a simple linear chain, allowing only limited variability in interaction dynamics.

When $d \geq 4$, there is an increase in possible topologies, and in the same way, the influence of these topologies on fire propagation. For $d = 4$, two distinct tree configurations exist, while for $d = 5$, there are three. The methodology of building the cdf of B is identical for both $d = 4$ and $d = 5$, that is why the detailed derivations for $d = 4$ are provided in [Appendix 2](#), whereas the case $d = 5$ is examined exhaustively in the present section.

For each tree configuration with 5 units, we explicitly derive, in what follows, the joint distribution of fire outcomes, examine how fire propagation probabilities influence the system's behavior, and compare the resulting losses using stochastic orderings such as the increasing convex order (IC order), also known as stop-loss order.

5.1 Stochastic representation of B depending on the structure of the site

Let $\mathcal{T}^{5,1} = (\mathcal{V}, \mathcal{E}_{5,1})$, $\mathcal{T}^{5,2} = (\mathcal{V}, \mathcal{E}_{5,2})$ and $\mathcal{T}^{5,3} = (\mathcal{V}, \mathcal{E}_{5,3})$ denote the series tree and hub trees depicted in [Figure 5](#) with corresponding set of edges $\mathcal{E}_{5,1} = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$, $\mathcal{E}_{5,2} = \{(1, 2), (2, 3), (3, 4), (3, 5)\}$, and $\mathcal{E}_{5,3} = \{(1, 3), (2, 3), (3, 4), (3, 5)\}$. For each structure, the pmfs $f_{B|\Theta=s}$ are given in [Tables 7, 9, and 11](#). The same pmfs for the case $d = 4$ in [Tables 5 and 6](#).

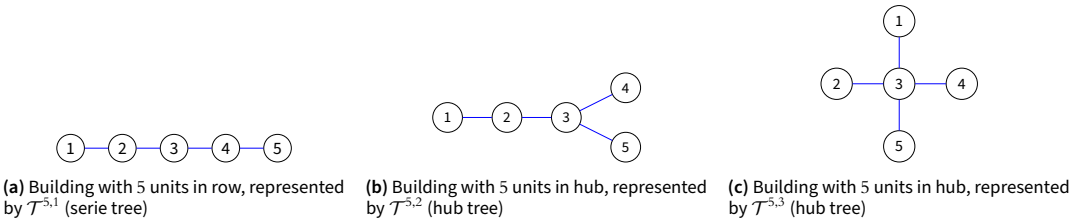


Figure 5. Different tree structures representing a 5-unit building

We focus, in what follows, on the structure $\mathcal{T}^{5,2}$ as results for the other two structures can be obtained in a similar way as $\mathcal{T}^{5,2}$ incorporates characteristics of both $\mathcal{T}^{5,1}$ and $\mathcal{T}^{5,3}$.

The stochastic representation of the rv $(B | \Theta = s)$, $s \in \mathcal{V}$, depends on the chosen tree configuration. To indicate the underlying tree, we use a superscript on B . The rv $(B^{\mathcal{T}^{5,2}} | \Theta = s)$ in [\(8\)](#) admits the

following stochastic representation:

$$(B^{T^{5,2}}|\Theta = s) = \begin{cases} (I_1|\Theta = 1) \left(Y_1 + (I_2|I_1 = 1)Y_2 + (I_2|I_1 = 1)(I_3|I_2 = 1)Y_3 + (I_2|I_1 = 1)(I_3|I_2 = 1)(I_4|I_3 = 1)Y_4 \right. \\ \quad \left. + (I_2|I_1 = 1)(I_3|I_2 = 1)(I_5|I_3 = 1)Y_5 \right), & s = 1, \\ (I_2|\Theta = 2) \left(Y_2 + (I_1|I_2 = 1)Y_1 + (I_3|I_2 = 1)Y_3 + (I_3|I_2 = 1)(I_4|I_3 = 1)Y_4 \right. \\ \quad \left. + (I_3|I_2 = 1)(I_5|I_3 = 1)Y_5 \right), & s = 2, \\ (I_3|\Theta = 3) \left(Y_3 + (I_2|I_3 = 1)Y_2 + (I_1|I_4 = 1)(I_4|I_3 = 1)Y_1 + (I_4|I_2 = 1)(I_2|I_3 = 1)Y_4 \right. \\ \quad \left. + (I_5|I_2 = 1)(I_2|I_3 = 1)Y_5 \right), & s = 3, \\ (I_4|\Theta = 4) \left(Y_4 + (I_3|I_4 = 1)Y_3 + (I_2|I_3 = 1)(I_3|I_4 = 1)Y_2 + (I_3|I_4 = 1)(I_2|I_3 = 1)(I_1|I_2 = 1)Y_1 \right. \\ \quad \left. + (I_5|I_3 = 1)(I_3|I_4 = 1)Y_5 \right), & s = 4, \\ (I_5|\Theta = 5) \left(Y_5 + (I_3|I_5 = 1)Y_3 + (I_2|I_3 = 1)(I_3|I_5 = 1)Y_2 + (I_3|I_5 = 1)(I_2|I_3 = 1)(I_1|I_2 = 1)Y_1 \right. \\ \quad \left. + (I_4|I_3 = 1)(I_3|I_5 = 1)Y_4 \right), & s = 5, \end{cases}$$

where $(I_{\nu_2}|I_{\nu_1} = 1)$ and $(I_{\nu_1}|I_{\nu_2} = 1)$ both follow a Bernoulli distribution of parameter $p_{\nu_1 \vee \nu_2}$ for $(\nu_1, \nu_2) \in \mathcal{E}_{5,2}$. Because the model assumes that fire can only spread from a burning vertex to an adjacent vertex (this means, there is no spontaneous ignition), it is interesting to note that $\Pr(I_{\nu_2} = 1|I_{\nu_1} = 0) = 0$ and $\Pr(I_{\nu_2} = 0|I_{\nu_1} = 0) = 1$, for $(\nu_1, \nu_2) \in \mathcal{E}_{5,2}$.

To simplify the notation, let us denote by $F_{Y_{i+j+\dots+k}}(x)$ the cdf of the sum $Y_i + Y_j + \dots + Y_k$. For instance, $F_{Y_{1+2+4}}(x)$ replaces $F_{Y_1+Y_2+Y_4}(x)$, and $F_{Y_{2+3}}(x)$ replaces $F_{Y_2+Y_3}(x)$. Based on (8), and using the expression of $B^{T^{5,2}}$ together with the strictly positive values of the joint pmf $f_{I|\Theta=s}$, $s \in \mathcal{V}$, given in Table 9 and the the values of $f_{B|\Theta=s}$, for $s \in \mathcal{V}$ given in Table 10, the expression of $F_{B|\Theta=s}$ in (10) becomes, for $x \in [0, b]$:

$$F_{B^{T^{5,2}}|\Theta=s}(x) = \begin{cases} p_1 \left(q_{1,2}F_{Y_1}(x) + p_{1,2}q_{2,3}F_{Y_{1+2}}(x) + p_{1,2}p_{2,3}q_{3,4}q_{3,5}F_{Y_{1+2+3}}(x) \right. \\ \quad \left. + p_{1,2}p_{2,3}p_{3,4}q_{3,5}F_{Y_{1+2+3+4}}(x) + p_{1,2}p_{2,3}q_{3,4}p_{3,5}F_{Y_{1+2+3+5}}(x) \right. \\ \quad \left. + p_{1,2}p_{2,3}p_{3,4}p_{3,5}F_{Y_{1+2+3+4+5}}(x) \right) + q_1, & s = 1, \\ p_2 \left(p_{2,1}q_{2,3}F_{Y_{1+2}}(x) + p_{2,1}p_{2,3}q_{3,4}q_{3,5}F_{Y_{1+2+3}}(x) \right. \\ \quad \left. + p_{2,1}p_{2,3}p_{3,4}q_{3,5}F_{Y_{1+2+3+4}}(x) + q_{2,1}q_{2,3}F_{Y_2}(x) \right. \\ \quad \left. + p_{2,1}p_{2,3}q_{3,4}p_{3,5}F_{Y_{1+2+3+5}}(x) + q_{2,1}p_{2,3}q_{3,4}q_{3,5}F_{Y_{2+3}}(x) \right. \\ \quad \left. + q_{2,1}p_{2,3}p_{3,4}q_{3,5}F_{Y_{2+3+4}}(x) + q_{1,2}p_{2,3}q_{3,4}p_{3,5}F_{Y_{2+3+5}}(x) \right. \\ \quad \left. + q_{2,1}p_{2,3}p_{3,4}p_{3,5}F_{Y_{2+3+4+5}}(x) + p_{2,1}p_{2,3}p_{3,4}p_{3,5}F_{Y_{1+2+3+4+5}}(x) \right) + q_2, & s = 2, \\ p_3 \left(p_{3,2}p_{2,1}q_{3,4}q_{3,5}F_{Y_{1+2+3}}(x) \right. \\ \quad \left. + p_{3,2}p_{2,1}p_{3,4}q_{3,5}F_{Y_{1+2+3+4}}(x) + q_{3,2}q_{3,4}q_{3,5}F_{Y_3}(x) \right. \\ \quad \left. + p_{3,2}p_{2,1}q_{3,4}p_{3,5}F_{Y_{1+2+3+5}}(x) + p_{3,2}q_{2,1}p_{3,4}p_{3,5}F_{Y_{2+3+4+5}}(x) \right. \\ \quad \left. + p_{3,2}q_{2,1}p_{3,4}q_{3,5}F_{Y_{2+3+4}}(x) + p_{3,2}q_{2,1}q_{3,4}p_{3,5}F_{Y_{2+3+5}}(x) \right. \\ \quad \left. + q_{3,2}p_{2,1}p_{3,4}q_{3,5}F_{Y_{3+4+5}}(x) + q_{3,2}p_{3,4}q_{3,5}F_{Y_{3+4}}(x) + p_{3,2}q_{2,1}q_{3,4}q_{3,5}F_{Y_{2+3}}(x) \right. \\ \quad \left. + q_{3,2}q_{3,4}p_{3,5}F_{Y_{3+5}}(x) + p_{2,1}p_{2,3}p_{3,4}p_{3,5}F_{Y_{1+2+3+4+5}}(x) \right) + q_3, & s = 3, \\ p_4 \left(q_{4,3}F_{Y_4}(x) + p_{4,3}q_{3,2}q_{3,5}F_{Y_{3+4}}(x) + p_{4,3}q_{3,2}p_{3,5}F_{Y_{3+4+5}}(x) \right. \\ \quad \left. + p_{4,3}p_{3,2}q_{2,1}p_{3,5}F_{Y_{2+3+4+5}}(x) + p_{4,3}p_{3,2}p_{2,1}q_{3,5}F_{Y_{1+2+3+4}}(x) \right. \\ \quad \left. + p_{4,3}p_{3,2}q_{2,1}q_{3,5}F_{Y_{2+3+4}}(x) + p_{4,3}p_{3,2}p_{2,1}p_{3,5}F_{Y_{1+2+3+4+5}}(x) \right) + q_4, & s = 4, \\ p_5 \left(q_{5,3}F_{Y_5}(x) + p_{5,3}q_{3,2}q_{3,4}F_{Y_{3+5}}(x) + p_{5,3}p_{3,2}q_{3,4}F_{Y_{3+4+5}}(x) \right. \\ \quad \left. + p_{5,3}p_{3,2}q_{2,1}p_{3,4}F_{Y_{2+3+4+5}}(x) + p_{5,3}p_{3,2}q_{3,4}p_{2,1}F_{Y_{1+2+3+4+5}}(x) \right. \\ \quad \left. + p_{5,3}q_{3,4}p_{3,2}q_{2,1}F_{Y_{2+3+5}}(x) + p_{5,3}p_{3,2}p_{3,4}p_{2,1}F_{Y_{1+2+3+4+5}}(x) \right) + q_5, & s = 5. \end{cases}$$

We can give in the same way the expressions of $F_{B^{T^{5,1}}|\Theta=s}(x)$ and $F_{B^{T^{5,3}}|\Theta=s}(x)$, using the values

of $f_{B|\Theta=s}$ in Table 8 and Table 12.

5.2 Impact of the propagation probabilities using the usual stochastic order

One of the key parameters of our model is the propagation probability within the structure. We now aim to examine how an increase (or a decrease) of this probability impacts the total losses incurred rv B based on the usual stochastic order. Such a situation may arise with lower (or greater) fire resistance materials added to separate different units of the site. We recall below the definition of the usual stochastic order (see e.g. Müller and Stoyan 2002) used in Definition 2. For more insight into stochastic ordering, see also Denuit et al. 2006 and Shaked and Shanthikumar 2007.

Definition 2 (Usual stochastic order). *Let X and X' be positive rvs with cdfs F_X and $F_{X'}$, respectively. We say that X is smaller than X' under the usual stochastic order, denoted $X \preceq_{st} X'$, if $F_X(x) \geq F_{X'}(x) \forall x \geq 0$.*

We use the usual stochastic order to qualify the impact of the propagation probabilities on the magnitude of the total losses incurred on a site when a fire occurs.

Theorem 1. *Consider the trees $\mathcal{T}^{5,1} = (\mathcal{V}, \mathcal{E}_{5,1})$ (serie tree), $\mathcal{T}^{5,2} = (\mathcal{V}, \mathcal{E}_{5,2})$ (hub tree 1), and $\mathcal{T}^{5,3} = (\mathcal{V}, \mathcal{E}_{5,3})$ (hub tree 2). Let B and B' be defined on the same tree $\mathcal{T}$ as in (8), where $\mathcal{T}$ is either $\mathcal{T}^{5,1}$, $\mathcal{T}^{5,2}$, or $\mathcal{T}^{5,3}$. If $0 < p_e \leq p'_e < 1$ and $Y_e \stackrel{\mathcal{D}}{=} Y'_e$ for $e \in \mathcal{E}$, then we have $B \preceq_{st} B'$.*

Proof. Without loss of generality, assume $\Theta = 1$. For the tree $\mathcal{T}^{5,2}$, rvs $(I_2|I_1 = 1)$, $(I_3|I_2 = 1)$, $(I_4|I_3 = 1)$, and $(I_5|I_4 = 1)$ are independent given the global Markov property stated in Definition 1. The Equation (9), shows that both $(B|\Theta = 1)$ and $(B'|\Theta' = 1)$ are increasing functions of the independent rvs $(I_2|I_1 = 1)$, $(I_3|I_2 = 1)$, $(I_4|I_3 = 1)$, $(I_5|I_4 = 1)$, Y_1, Y_2, Y_3, Y_4, Y_5 for $(B|\Theta = 1)$ and of $(I'_2|I'_1 = 1)$, $(I'_3|I'_2 = 1)$, $(I'_4|I'_3 = 1)$, $(I'_5|I'_4 = 1)$, $Y'_1, Y'_2, Y'_3, Y'_4, Y'_5$ for $(B'|\Theta' = 1)$.

By assumption from the theorem statement, $(I_2|I_1 = 1) \preceq_{st} (I'_2|I'_1 = 1)$, $(I_3|I_2 = 1) \preceq_{st} (I'_3|I'_2 = 1)$, $(I_4|I_3 = 1) \preceq_{st} (I'_4|I'_3 = 1)$, and $(I_5|I_4 = 1) \preceq_{st} (I'_5|I'_4 = 1)$ while $Y_e \stackrel{\mathcal{D}}{=} Y'_e$ for $e \in \mathcal{E}$. Applying Theorem 1.2.16 of Müller and Stoyan 2002, we find that $(B|\Theta = 1) \preceq_{st} (B'|\Theta' = 1)$. We repeat the same rationale for $j \in \{2, 3, 4, 5\}$. From (10) and having shown that $(B|\Theta = s) \preceq_{st} (B'|\Theta' = s)$ for $s \in \mathcal{V}$, we find that

$$F_B(x) = \sum_{s \in \mathcal{V}} f_{\Theta}(s) F_{B|\Theta=s}(x) = \sum_{s \in \mathcal{V}} f_{\Theta'}(s) F_{B|\Theta=s}(x) \geq \sum_{s \in \mathcal{V}} f_{\Theta'}(s) F_{B'|\Theta'=s}(x) = F_{B'}(x), \quad \forall x \in [0, b],$$

implying the desired result $B \preceq_{st} B'$. For B and B' defined on the same tree $\mathcal{T}^{5,1}$ (or $\mathcal{T}^{5,3}$), the proof is similar. $\square$

Theorem 1 coupled with the property of the usual stochastic order has important implications. First, if the conditions of Theorem 1 are satisfied, it means that $E[\phi(B)] \leq E[\phi(B')]$, for all increasing integrable functions ϕ , with the consequence that $E[B] \leq E[B']$. Knowing that the expectation of B is the pure premium of B in non-life, it highlights the significant implications of Theorem 1 in the context of non-life insurance pricing.

We recall the definitions of the Value-at-Risk (VaR) and Tail-Value-at-Risk (TVaR, also known as expected shortfall) of B at confidence level κ given by $\text{VaR}_{\kappa}(B) = F_B^{-1}(\kappa)$ and $\text{TVaR}_{\kappa}(B) = \frac{1}{1-\kappa} \int_{\kappa}^1 F_B^{-1}(u) du$, for $\kappa \in (0, 1)$ (see, Denuit et al. 2006 and McNeil, Frey, and Embrechts 2015 for details on those two risk measures and their properties).

Given that $B \preceq_{st} B'$ and $F_B^{-1}(\kappa) \leq F_{B'}^{-1}(\kappa)$ for all $\kappa \in (0, 1)$, are equivalent, it follows, for the VaR and TVaR measures, that the relations $\text{VaR}_{\kappa}(B) \leq \text{VaR}_{\kappa}(B')$ and $\text{TVaR}_{\kappa}(B) \leq \text{TVaR}_{\kappa}(B')$ hold for all $\kappa \in (0, 1)$, see Section 3.3 of Denuit et al. 2006. Considering the entropic risk measure with the expression $\varphi_{\eta}(B) = \frac{1}{\eta} \ln(E[e^{\eta B}])$, $\eta > 0$, since $e^{\gamma x}$ is an increasing function for $\gamma > 0$, the inequality

$\varphi_\gamma(B) \leq \varphi_\gamma(B')$ is true for all $\gamma > 0$. Hence, Theorem 1 is crucial for application in at least two contexts: pricing in actuarial science and risk management (civil engineering). In the field of civil engineering, its application could involve reinforcing fire safety measures at the site, such as adding fire doors or sprinklers. These enhancements would reduce the probability of fire propagation and, consequently, decrease the magnitude of losses due to fire.

Example 1 (Illustration of the impact of propagation probabilities). We consider trees $\mathcal{T}^{5,1}$, $\mathcal{T}^{5,2}$ and $\mathcal{T}^{5,3}$ with identical probabilities of propagation $p_{v_1, v_2} = p$ along all edges for $(v_1, v_2) \in \mathcal{E}^{5,m}$, $m = 1, 2, 3$. We set each unit $s \in \mathcal{V}$, the damage amount to follow a discretized beta distribution with pmf $f_{Y_s}(k) = \left(\frac{k}{c_s}\right)^{\alpha_s} - \left(\frac{k-1}{c_s}\right)^{\alpha_s}$, $k \in \{1, \dots, c_s\}$, $s \in \mathcal{V}$, and parameters summarized in Table 1a.

Table 1. Model parameters and loss statistics

Unit s	Description	α_s	c_s	$f_\Theta(s)$	$\Pr(I_s = 1 \mid \Theta = s)$
1	Bedroom	2	100	0.1	0.2
2	Office	1.5	80	0.1	0.2
3	Kitchen	3	150	0.55	0.4
4	Living room	2.5	120	0.2	0.2
5	Bathroom	1	50	0.05	0.1

(a) Parameters settings

From	To (Separated by)	$p_{i \rightarrow j}$
Room	Room (Wall)	0.81
Room	Room (Corridor)	0.84
Room	Shaft/Stairwell (Wall)	0.57
Room	Shaft/Stairwell (Corridor)	0.28
Shaft/Stairwell	Room (Wall)	0.60

(b) Fire propagation probabilities between different spaces, adapted from Cheng and Hadjisophocleous 2009.

The total loss variable B thus takes values in $\{1, \dots, b\}$, with $b = c_1 + \dots + c_5 = 500$. These assumptions are chosen for illustration, with probabilities of fire ignition (f_Θ) inspired by those in Section 3.2 of Li, Zhang, and Hadjisophocleous 2013. We prove through this that we are able to compute all desired values. In practice, parameters would be elicited from experts, such as engineers or underwriters. Some examples of probabilities of propagation from engineers are summarized in Table 1b.

The values are reported in Figure 6 and they illustrated the application of Lemma 1. In Table 2 and Table 3, we present the values of $E[B^{\mathcal{T}^{5,m}}]$, $m \in \{1, 2, 3\}$, for $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. As expected from Proposition 1, we observe that $E[B^{\mathcal{T}^{5,1}}] \leq E[B^{\mathcal{T}^{5,2}}] \leq E[B^{\mathcal{T}^{5,3}}]$, for all $p \in (0, 1)$.

Table 2. Values of $E[B^{\mathcal{T}^{5,1}}]$, $E[B^{\mathcal{T}^{5,2}}]$, $E[B^{\mathcal{T}^{5,3}}]$, $VaR_{0.9}(B^{\mathcal{T}^{5,1}})$, $VaR_{0.9}(B^{\mathcal{T}^{5,2}})$ and $VaR_{0.9}(B^{\mathcal{T}^{5,3}})$, for propagation probability $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$

p	$E[B^{\mathcal{T}^{5,1}}]$	$VaR_{0.9}(B^{\mathcal{T}^{5,1}})$	$E[B^{\mathcal{T}^{5,2}}]$	$VaR_{0.9}(B^{\mathcal{T}^{5,2}})$	$E[B^{\mathcal{T}^{5,3}}]$	$VaR_{0.9}(B^{\mathcal{T}^{5,3}})$
0.9	93.00	348.00	93.55	348.00	95.30	349.00
0.7	74.02	290.00	75.21	294.00	79.07	306.00
0.5	58.26	233.00	59.58	237.00	63.92	255.00
0.3	45.37	171.00	46.42	178.00	49.85	199.00
0.1	35.04	133.00	35.47	135.00	36.85	139.00

Table 3. Values of $\varphi_{0.01}(B^{\mathcal{T}^{5,1}})$, $\varphi_{0.01}(B^{\mathcal{T}^{5,2}})$ and $\varphi_{0.01}(B^{\mathcal{T}^{5,3}})$, for propagation probability $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$

p	$TVaR_{0.9}(B^{\mathcal{T}^{5,1}})$	$TVaR_{0.9}(B^{\mathcal{T}^{5,2}})$	$TVaR_{0.9}(B^{\mathcal{T}^{5,3}})$	$\varphi_{0.01}(B^{\mathcal{T}^{5,1}})$	$\varphi_{0.01}(B^{\mathcal{T}^{5,2}})$	$\varphi_{0.01}(B^{\mathcal{T}^{5,3}})$
0.9	383.84	383.85	384.23	219.74	220.50	223.22
0.7	345.64	346.70	351.74	176.33	178.34	186.01
0.5	292.02	294.84	308.30	132.83	135.49	146.32
0.3	234.34	237.53	253.79	92.34	94.74	105.04
0.1	167.09	169.89	180.51	58.90	59.97	64.47

In Figure 6 we present the cdfs of losses for trees $B^{\mathcal{T}^{5,m}}$, $m = 1, 2, 3$ for different values of probabilities of propagation to highlight the impact of increasing or decreasing the value of these probabilities. On that figure, all values are exact (there is no simulation). We computed the values using probability generating functions and the FFT algorithm. We can observe that the Expectation and VaR of the claim amount rvs $B^{\mathcal{T}^{5,1}}$, $B^{\mathcal{T}^{5,2}}$ and $B^{\mathcal{T}^{5,3}}$ decrease as propagation probabilities decrease. These observations are confirmed by Theorem 1.

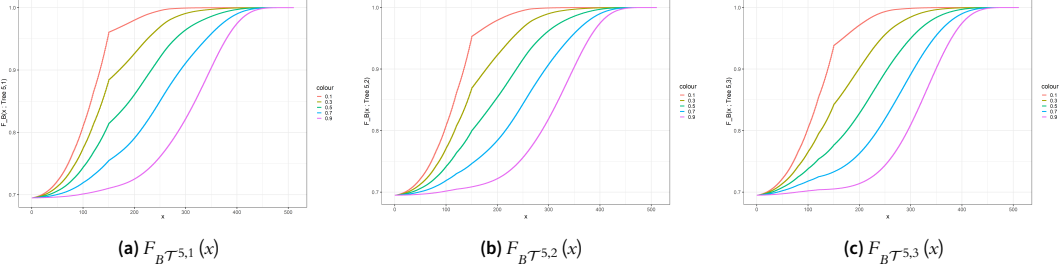


Figure 6. $F_{B^{\mathcal{T}^{5,m}}}(x)$, $m \in \{1, 2, 3\}$ and $p = 0.1, 0.3, 0.5, 0.7, 0.9$

5.3 Marginal distribution of the components of I

We investigate the marginal distributions of the components of I . At unit $\nu \in \mathcal{V}$, the rv I_ν follows a Bernoulli distribution with parameter p_ν^{bern} which can be decomposed in two components such as

$$p_\nu^{\text{bern}} = f_\Theta(\nu)p_\nu + \gamma_\nu, \quad \nu \in \mathcal{V},$$

where $f_\Theta(\nu)$ is the probability that the fire ignites at unit ν and γ_ν , the probability that the fire reaches unit ν , $\nu \in \mathcal{V}$ when it ignites at another unit. For the tree $\mathcal{T}^{5,2} = (\mathcal{V}, \mathcal{E}_{5,2})$, the component γ_ν , using $f_{I|\Theta=s}$ of Table 11, is given by

$$\gamma_\nu^{\mathcal{T}^{5,2}} = \begin{cases} f_\Theta(2)p_2p_{2 \rightarrow 1} + f_\Theta(3)p_3p_{3 \rightarrow 2}p_{2 \rightarrow 1} + f_\Theta(4)p_4p_{4 \rightarrow 3}p_{3 \rightarrow 2}p_{2 \rightarrow 1} + f_\Theta(5)p_5p_{5 \rightarrow 3}p_{3 \rightarrow 2}p_{2 \rightarrow 1}, & \nu = 1, \\ f_\Theta(1)p_1p_{1 \rightarrow 2} + f_\Theta(3)p_3p_{3 \rightarrow 2} + f_\Theta(4)p_4p_{4 \rightarrow 3}p_{3 \rightarrow 2} + f_\Theta(5)p_5p_{5 \rightarrow 3}p_{3 \rightarrow 2}, & \nu = 2, \\ f_\Theta(1)p_1p_{1 \rightarrow 2}p_{2 \rightarrow 3} + f_\Theta(2)p_2p_{2 \rightarrow 3} + f_\Theta(4)p_4p_{4 \rightarrow 3} + f_\Theta(5)p_5p_{5 \rightarrow 3}, & \nu = 3, \\ f_\Theta(1)p_1p_{1 \rightarrow 2}p_{2 \rightarrow 3}p_{3 \rightarrow 4} + f_\Theta(2)p_2p_{2 \rightarrow 3}p_{3 \rightarrow 4} + f_\Theta(3)p_3p_{3 \rightarrow 4} + f_\Theta(5)p_5p_{5 \rightarrow 3}p_{3 \rightarrow 4}, & \nu = 4, \\ f_\Theta(1)p_1p_{1 \rightarrow 2}p_{2 \rightarrow 3}p_{3 \rightarrow 5} + f_\Theta(2)p_2p_{2 \rightarrow 3}p_{3 \rightarrow 5} + f_\Theta(3)p_3p_{3 \rightarrow 5} + f_\Theta(4)p_4p_{4 \rightarrow 3}p_{3 \rightarrow 5}, & \nu = 5. \end{cases}$$

The probability γ_ν is an increasing function of the propagation probabilities. If connections between all units of the site are such that $p_{\nu_1 \rightarrow \nu_2} = 0$, for all $(\nu_1, \nu_2) \in \mathcal{E}_{4,1}$ or $(\nu_1, \nu_2) \in \mathcal{E}_{4,2}$, then γ_ν is equal to 0. On the contrary, $\gamma_\nu^{\mathcal{T}^{5,2}} = \sum_{j \neq \nu} \prod_{(a,b) \in \text{path}(j \rightarrow \nu)} p_{a \rightarrow b} f_\Theta(j) p_j$, if the connections between all units

of the site are such that $p_{\nu_1 \rightarrow \nu_2} = 1$, for all $(\nu_1, \nu_2) \in \mathcal{E}_{5,2}$. So, efforts to mitigate fire risk should focus not only on reducing the probability of fire initiation but also on managing the connectivity and propagation pathways between units. Introducing fire barriers or materials with high fire resistance between key units could effectively lower γ_s , thereby reducing the overall risk.

Within the settings of Theorem 1 and assuming $\mathcal{T} = \mathcal{T}^{5,2}$, we define γ'_ν by replacing $p_{\nu_1 \rightarrow \nu_2}$ by $p'_{\nu_1 \rightarrow \nu_2}$ in above Equation for all $(\nu_1, \nu_2) \in \mathcal{E}_{5,2}$.

If $0 < p_{\nu_1 \rightarrow \nu_2} \leq p'_{\nu_1 \rightarrow \nu_2} < 1$ for all $(\nu_1, \nu_2) \in \mathcal{E}_{5,2}$, then $\gamma_\nu \leq \gamma'_\nu$, for $\nu \in \mathcal{V}$. Hence, increasing the probabilities of fire propagation between units leads to a higher or equal probability γ_s that the fire

will reach unit ν , $\nu \in \mathcal{V}$, when it starts at another unit. We can say that there is a need to control these inter-unit probabilities of propagation to mitigate the overall risk to each unit. This is true since higher connectivity in terms of fire propagation makes each unit more vulnerable to being affected by a fire that originates elsewhere in the structure.

We highlight the fact that the inequality $\gamma_\nu \leq \gamma'_\nu$ aligns with the result of Theorem 1, which establishes that $B \preceq_{st} B'$. The increase in probabilities of propagation not only raises γ_ν , the likelihood that a fire reaches unit ν , but also reflects a broader increase in the overall risk as captured by the stochastic order between B and B' . This indicates that the risk of fire loss increases in a manner consistent with the increase in individual probabilities of propagation.

5.4 Comparison of trees based on the IC order

We now investigate how the topology of the structure $\mathcal{T}$ influences the distribution of the total loss random variable B . To do so, we define, for each structure $\mathcal{T}^{5,m}$, $m \in \{1, 2, 3\}$, the random variables representing the number of fire incidents $M^{\mathcal{T}^{5,m}}$ and the occurrence of fires in each unit $I_\nu^{\mathcal{T}^{5,m}}$, $\nu \in \mathcal{V}$. We introduce the following additional assumptions:

- A1 Identical propagation probabilities: $p_e^{\mathcal{T}^{5,1}} = p$ for $e \in \mathcal{E}_{5,1}$, $p_e^{\mathcal{T}^{5,2}} = p$ for $e \in \mathcal{E}_{5,2}$, and $p_e^{\mathcal{T}^{5,3}} = p$ for $e \in \mathcal{E}_{5,3}$;
- A2 Identically distributed loss damage rvs: $Y_s^{\mathcal{T}^{5,1}} \stackrel{\mathcal{D}}{=} Y_s^{\mathcal{T}^{5,2}} \stackrel{\mathcal{D}}{=} Y_s^{\mathcal{T}^{5,3}} \stackrel{\mathcal{D}}{=} Y$, for $s \in \mathcal{V}$, where the cdf of the positive rv Y is denoted by F_Y ;
- A3 Discrete uniform distribution: $\int_{\Theta^{\mathcal{T}^{5,1}}}(s) = \int_{\Theta^{\mathcal{T}^{5,2}}}(s) = \int_{\Theta^{\mathcal{T}^{5,3}}}(s) = \frac{1}{5}$, $s \in \mathcal{V}$;
- A4 Fires always flashover in a unit when it starts within that unit $p_1 = p_2 = p_3 = p_4 = p_5 = 1$.

We define the rvs

$$M^{\mathcal{T}^{5,1}} = \sum_{\nu \in \mathcal{V}} I_\nu^{\mathcal{T}^{5,1}}, \quad M^{\mathcal{T}^{5,2}} = \sum_{\nu \in \mathcal{V}} I_\nu^{\mathcal{T}^{5,2}} \quad \text{and} \quad M^{\mathcal{T}^{5,3}} = \sum_{\nu \in \mathcal{V}} I_\nu^{\mathcal{T}^{5,3}}, \quad (11)$$

with cdfs respectively given by

$$F_{M^{\mathcal{T}^{5,1}}}(x) = \begin{cases} \frac{1}{5} (5 - 8p + 3p^2) & , 1 \leq x < 2, \\ \frac{1}{5} (5 - 9p^2 + 4p^3) & , 2 \leq x < 3, \\ \frac{1}{5} (5 - 8p^3 + 3p^4) & , 3 \leq x < 4, \end{cases} \quad F_{M^{\mathcal{T}^{5,2}}}(x) = \begin{cases} \frac{1}{5} (5 - 8p + 4p^2 - p^3) & , 1 \leq x < 2, \\ \frac{1}{5} (5 - 12p^2 + 9p^3 - 2p^4) & , 2 \leq x < 3, \\ \frac{1}{5} (5 - 12p^3 + 7p^4) & , 3 \leq x < 4, \end{cases}$$

and

$$F_{M^{\mathcal{T}^{5,3}}}(x) = \begin{cases} \frac{1}{5} (5 - 8p + 6p^2 - 4p^3 + p^4) & , 1 \leq x < 2, \\ \frac{1}{5} (5 - 18p^2 + 20p^3 - 7p^4) & , 2 \leq x < 3, \\ \frac{1}{5} (5 - 16p^3 + 11p^4) & , 3 \leq x < 4, \end{cases}$$

with $F_{M^{\mathcal{T}^{5,1}}}(x) = F_{M^{\mathcal{T}^{5,2}}}(x) = F_{M^{\mathcal{T}^{5,3}}}(x) = 0$ for $0 \leq x < 1$, $F_{M^{\mathcal{T}^{5,1}}}(x) = F_{M^{\mathcal{T}^{5,2}}}(x) = F_{M^{\mathcal{T}^{5,3}}}(x) = 1 - p^4$ for $4 \leq x < 5$ and $F_{M^{\mathcal{T}^{5,1}}}(x) = F_{M^{\mathcal{T}^{5,2}}}(x) = F_{M^{\mathcal{T}^{5,3}}}(x) = 1$ for $x \geq 5$.

The cdfs of $B^{\mathcal{T}^{5,1}}$, $B^{\mathcal{T}^{5,2}}$, and $B^{\mathcal{T}^{5,3}}$ become

$$\begin{aligned} F_{B^{\mathcal{T}^{5,1}}}(x) &= \frac{1}{5} (2q + 3q^2) F_Y(x) + \frac{4}{5} (pq + pq^2) F_Y^{*2}(x) \\ &\quad + \frac{3}{5} (2p^2q + p^2q^2) F_Y^{*3}(x) + \frac{8}{5} p^3 q F_Y^{*4}(x) + p^4 F_Y^{*5}(x), \\ F_{B^{\mathcal{T}^{5,2}}}(x) &= \frac{1}{5} (3q + q^2 + q^3) F_Y(x) + \frac{2}{5} (pq + pq^3 + 2pq^2) F_Y^{*2}(x) \\ &\quad + \frac{3}{5} (p^2q + 3p^2q^2) F_Y^{*3}(x) + \frac{12}{5} p^3 q F_Y^{*4}(x) + p^4 F_Y^{*5}(x), \\ F_{B^{\mathcal{T}^{5,3}}}(x) &= \frac{1}{5} (q^4 + 4q) F_Y(x) + \frac{8}{5} pq^3 F_Y^{*2}(x) \\ &\quad + \frac{18}{5} p^2 q^2 F_Y^{*3}(x) + \frac{16}{5} p^3 q F_Y^{*4}(x) + p^4 F_Y^{*5}(x), \end{aligned}$$

for $x \in [0, b]$. Here, $F_Y^{*k}(x)$ denotes the k -fold convolution of F_Y , this means $F_Y^{*k}(x) = \Pr(Y_1 + \dots + Y_k \leq x)$, where $Y_1, \dots, Y_k$ are i.i.d. with cdf F_Y . We can now compare $B^{\mathcal{T}^{5,1}}$, $B^{\mathcal{T}^{5,2}}$ and $B^{\mathcal{T}^{5,3}}$ under the IC order.

Definition 3 (IC order). *Let X and X' be positive random variables with cdfs F_X and $F_{X'}$, respectively. Assume that $E[X] \leq E[X'] < \infty$. We say that X is smaller than X' under the IC order, denoted $X \preceq_{sl} X'$, if $E[\max(X - x; 0)] \leq E[\max(X' - x; 0)]$ is satisfied for all $x \geq 0$.*

The Karlin-Novikoff Criterion provided in the following lemma is useful for establishing the IC order between two rvs.

Lemma 1 (Karlin-Novikoff Criterion). *Let X and X' be positive random variables with cdfs F_X and $F_{X'}$, respectively. Assume that $E[X] \leq E[X'] < \infty$. We say that X is smaller than X' under the IC order, denoted $X \preceq_{sl} X'$, if there exists a number c such that $F_X(x) \leq F_{X'}(x)$ for $x < c$ and $F_X(x) \geq F_{X'}(x)$ for $x \geq c$, for all $x \in \mathbb{R}$.*

The desired result is provided in the following proposition.

Proposition 1. *Let the rvs $M^{\mathcal{T}^{5,1}}$, $M^{\mathcal{T}^{5,2}}$, and $M^{\mathcal{T}^{5,3}}$ be defined as in (11). Let $B^{\mathcal{T}^{5,1}}$, $B^{\mathcal{T}^{5,2}}$ and $B^{\mathcal{T}^{5,3}}$ be the corresponding loss damages rvs. Then, we have*

$$M^{\mathcal{T}^{5,1}} \preceq_{sl} M^{\mathcal{T}^{5,2}} \preceq_{sl} M^{\mathcal{T}^{5,3}}$$

and

$$B^{\mathcal{T}^{5,1}} \preceq_{sl} B^{\mathcal{T}^{5,2}} \preceq_{sl} B^{\mathcal{T}^{5,3}}.$$

Proof. For $M^{\mathcal{T}^{5,2}}$ versus $M^{\mathcal{T}^{5,1}}$, we have

$$F_{M^{\mathcal{T}^{5,2}}}(x) - F_{M^{\mathcal{T}^{5,1}}}(x) = \begin{cases} \frac{1}{5} p^2 (1 - p) > 0, & 1 \leq x < 2, \\ \frac{1}{5} p^2 ((p - 1) - 2p^2) < 0, & 2 \leq x < 3, \\ \frac{4}{5} p^3 (p - 1) < 0, & 3 \leq x < 4. \end{cases}$$

For the case $M^{\mathcal{T}^{5,3}}$ versus $M^{\mathcal{T}^{5,2}}$, we have

$$F_{M^{\mathcal{T}^{5,3}}}(x) - F_{M^{\mathcal{T}^{5,2}}}(x) = \begin{cases} \frac{1}{5} p^2 (2(1 - p) + p^2) > 0, & 1 \leq x < 2, \\ \frac{1}{5} p^2 (p(-5p + 11) - 6) < 0, & 2 \leq x < 3, \\ \frac{4}{5} p^3 (p - 1) < 0, & 3 \leq x < 4. \end{cases}$$

The results follows by applying lemma 1 with $F_{M^{\mathcal{T}^{5,2}}}(x) - F_{M^{\mathcal{T}^{5,1}}}(x)$ and $F_{M^{\mathcal{T}^{5,3}}}(x) - F_{M^{\mathcal{T}^{5,2}}}(x)$. Each $B^{\mathcal{T}^{5,i}}$ $i \in \{1, 2, 3\}$ can be represented as the sum of a random number $M^{\mathcal{T}^{5,i}}$ of independent and identically distributed severities, independent of $M^{\mathcal{T}^{5,i}}$. Since we have established that $M^{\mathcal{T}^{5,1}} \preceq_{sl} M^{\mathcal{T}^{5,2}} \preceq_{sl} M^{\mathcal{T}^{5,3}}$, applying Property 3.4.39 of Denuit et al. 2006 iteratively yields $B^{\mathcal{T}^{5,1}} \preceq_{sl} B^{\mathcal{T}^{5,2}} \preceq_{sl} B^{\mathcal{T}^{5,3}}$. $\square$

It is important to specify that the tree $\mathcal{T}^{5,2}$ is an intermediate structure, between the series tree ($\mathcal{T}^{5,1}$) and hub tree ($\mathcal{T}^{5,3}$). It effectively balances the characteristics of both extremes. This is illustrated in Example 2 below.

Example 2. We compute the values of $F_{M^{\mathcal{T}^{5,1}}}(x)$, $F_{M^{\mathcal{T}^{5,2}}}(x)$, and $F_{M^{\mathcal{T}^{5,3}}}(x)$ for $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The results, displayed in Figure 7, illustrate the application of Proposition 1. Table 4 summarizes the corresponding expected values, $\text{TVaR}_{0.9}$, and entropic risk measures. In line with Proposition 1, we consistently find

$$E[M^{\mathcal{T}^{5,1}}] \leq E[M^{\mathcal{T}^{5,2}}] \leq E[M^{\mathcal{T}^{5,3}}],$$

and the series configuration yields lower or equal risk measures compared to the hub configurations for all $p \in (0, 1)$.

Table 4. Summary of expected values for $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ assuming that $\Pr(Y_s = 1) = 1$, for all $s \in \mathcal{V}$

p	$E[M^{\mathcal{T}^{5,1}}]$	$E[M^{\mathcal{T}^{5,2}}]$	$E[M^{\mathcal{T}^{5,3}}]$	$\text{TVaR}_{0.9}(M^{\mathcal{T}^{5,1}})$	$\text{TVaR}_{0.9}(M^{\mathcal{T}^{5,2}})$	$\text{TVaR}_{0.9}(M^{\mathcal{T}^{5,3}})$
0.1	1.1730	1.1770	1.1840	2.1884	2.2460	2.3522
0.3	1.6130	1.6460	1.6960	3.4644	3.6156	3.7668
0.5	2.2250	2.3000	2.4000	4.6250	4.6250	4.6250
0.7	3.0780	3.1780	3.2960	5.0000	5.0000	5.0000
0.9	4.2580	4.3190	4.3840	5.0000	5.0000	5.0000

Additional comments

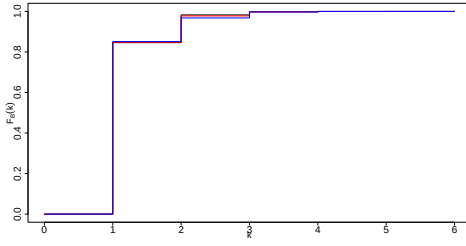
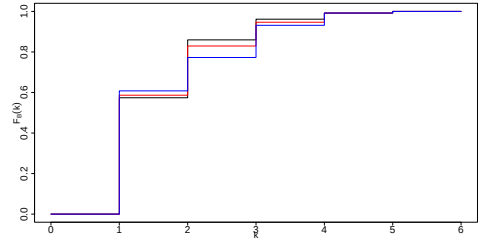
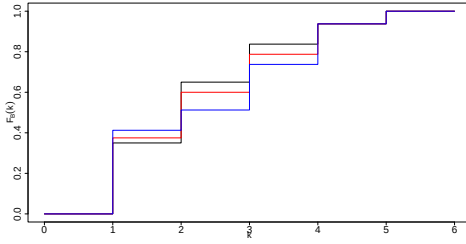
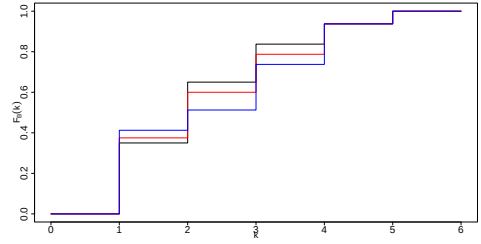
When the number of d units exceeds 5, the enumeration of all cases becomes tedious. In Chuisseau et al. 2025, we introduce within the actuarial fire spread model, a recursive method using both results from graph theory and the FFT algorithm that allows us to derive the cdf of B efficiently.

6. Pricing and Maxwell Boltzmann Bose Einstein Fermi Dirac (MBBEFD) Distribution

6.1 Introduction

The (MBBEFD) distribution is a flexible parametric family inspired initially by statistical mechanics and introduced by Bernegger 1997 for modeling property losses. It is designed to capture damage as a proportion of the maximum possible loss (MPL). This makes it particularly well-suited for applications in property insurance, where loss shapes vary depending on the type of construction or layout. Its parsimony with only two parameters is required, combined with its ability to flexibly approximate a variety of loss distributions, making the MBBEFD an attractive modeling tool in actuarial science.

The work of Parodi and Watson 2019 has strengthened the relevance of the MBBEFD distribution in fire risk modeling, which applied it in a graph-theoretic framework to describe the propagation of fire and explosions in industrial sites. In that same work, Parodi and Watson 2019 demonstrated that MBBEFD curves can effectively approximate the loss distribution resulting from fire propagation on graphs, where the distribution's shape depends on the graph's structure and the fire ignition

(a) For a propagation probability $p = 0.1$ (b) For a propagation probability $p = 0.3$ (c) For a propagation probability $p = 0.5$ (d) For a propagation probability $p = 0.7$ **Figure 7.** cdfs of losses, under the assumption that $\Pr(Y_s = 1) = 1$ for all $s \in \mathcal{V}$. $F_{BT^{5,1}}$ (black) $F_{BT^{5,2}}$ (red) $F_{BT^{5,3}}$ (Blue)

scenario. This modeling perspective has gained attraction in reinsurance as well, where the MBBEFD distribution is commonly used to derive exposure curves and severity distributions, especially in the context of fire, industrial, and catastrophe risk.

6.2 Mathematical definition

Mathematically, the cdf of a normalized loss $Z \in [0, 1]$ under the MBBEFD model is given by the following expression, as presented in Parodi and Watson 2019:

$$F_Z(x) = \begin{cases} \frac{\beta(\gamma - 1)(1 - \beta^x)}{\beta(\gamma - 1) + (1 - \beta\gamma)b^x}, & \text{if } 0 \leq x < 1, \\ 1, & \text{if } x = 1, \end{cases}$$

where $\beta > 0$ and $\gamma > 1$ are the distribution's parameters. The function F_Z is continuous on $[0, 1]$ and strictly increasing on $(0, 1)$ under typical parameter values. The corresponding density function $f_Z(x)$ exists and can be evaluated numerically, with $f_Z(x) = \frac{d}{dx}F_Z(x)$.

6.3 Model extension: Accounting for a mass at zero

In the context of our AFSM, there is a significant probability that no damage will occur. In that case, the random variable B representing the loss amount exhibits a considerable probability mass at zero. Elsewhere, the classical MBBEFD distribution is continuous and does not account for zero-loss events. So we extend the model to include a discrete mass at zero. To address this problem, we introduce a mixed model that combines both discrete and continuous components. Specifically, we

define a random variable W as

$$W = \begin{cases} Z, & \text{if } K = 1, \\ 0, & \text{if } K = 0, \end{cases}$$

where the binary rv K , indicates whether a loss occurs; $K \sim \text{Bernoulli}(\delta)$, and Z represents the normalized loss amount conditional on damage $Z \sim \text{MBBEFD}(\beta, \gamma)$. The parameter δ reflects the probability of observing a non-zero loss, and the complementary probability $1 - \delta$ corresponds to the probability of no loss occurring. In practice, we estimate $1 - \delta$ using the value of the model-based cdf of B at zero, that means, $1 - \delta = F_B(0)$. More precision about this parametrization can be found in Parodi and Watson 2019

The cdf of the resulting mixed random variable W takes the form:

$$F_W(x) = \begin{cases} 1 - \delta, & \text{if } x = 0, \\ 1 - \delta + \delta \times F_Z\left(\frac{x}{b}\right), & \text{if } 0 < x \leq b, \end{cases}$$

where b is the total insured value of the property, also called the maximum possible loss in our modeling context. We see that the formulation of $F_W(x)$ allows for a natural interpretation: the term $1 - \delta$ captures the frequency of zero-loss events, while the scaled MBBEFD component governs the distribution of positive losses. In terms of the original random variable B , the overall cdf can

be written as: $F_B(x) = \begin{cases} 1 - \delta, & \text{if } x = 0, \\ 1 - \delta + \delta \times F_{B'}(x), & \text{if } 0 < x \leq b, \end{cases}$ where $F_{B'}(x) = F_{B|B>0}(x)$ represents the conditional distribution of B given that a loss has occurred.

6.4 Estimation methodology

To fit the MBBEFD distribution to the analytically derived distribution of B from our AFSM, we consider two estimation approaches, as proposed in Parodi and Watson 2019. The first approach is based on maximum likelihood estimation (mle), where we compute the exact theoretical cdf of normalized losses and evaluate it at discrete points, typically ranging from 0 to 500. We estimate the parameters (β, γ) by maximizing the corresponding likelihood. The second approach that we used is based on least squares estimation (lse), which consists of minimizing the squared distance between the empirical cdf of the loss variable and the theoretical cdf F_W derived from the mixed model.

We adopt a reparameterization of the MBBEFD model (also suggested in Parodi and Watson 2019), which we use to improve numerical stability and facilitate optimization. So we express the parameters as $\beta = e^k$ and $\gamma = 1 + e^\ell$, which maps the parameter space to more manageable intervals, typically $k \in [-15, 5]$ and $\ell \in [-5, 20]$. This reparameterization reduces the likelihood surface's sensitivity to extreme values, enhances convergence during estimation, and improves the robustness of the fitting procedure.

Overall, we noticed that the MBBEFD distribution and its mixed extension provide a compact, flexible, and interpretable approach for approximating loss distributions derived from structural fire-spreading models. This methodology is especially advantageous when the available data are limited but a well-defined structural layout of the building or site is known.

6.5 Numerical application

To fit computed cdfs with MBBEFD models, we consider the tree structure $\mathcal{T}^{5,2}, (\nu_1, \nu_2) \in \mathcal{E}^{5,2}$, with identical parameters settings as in Example 1.

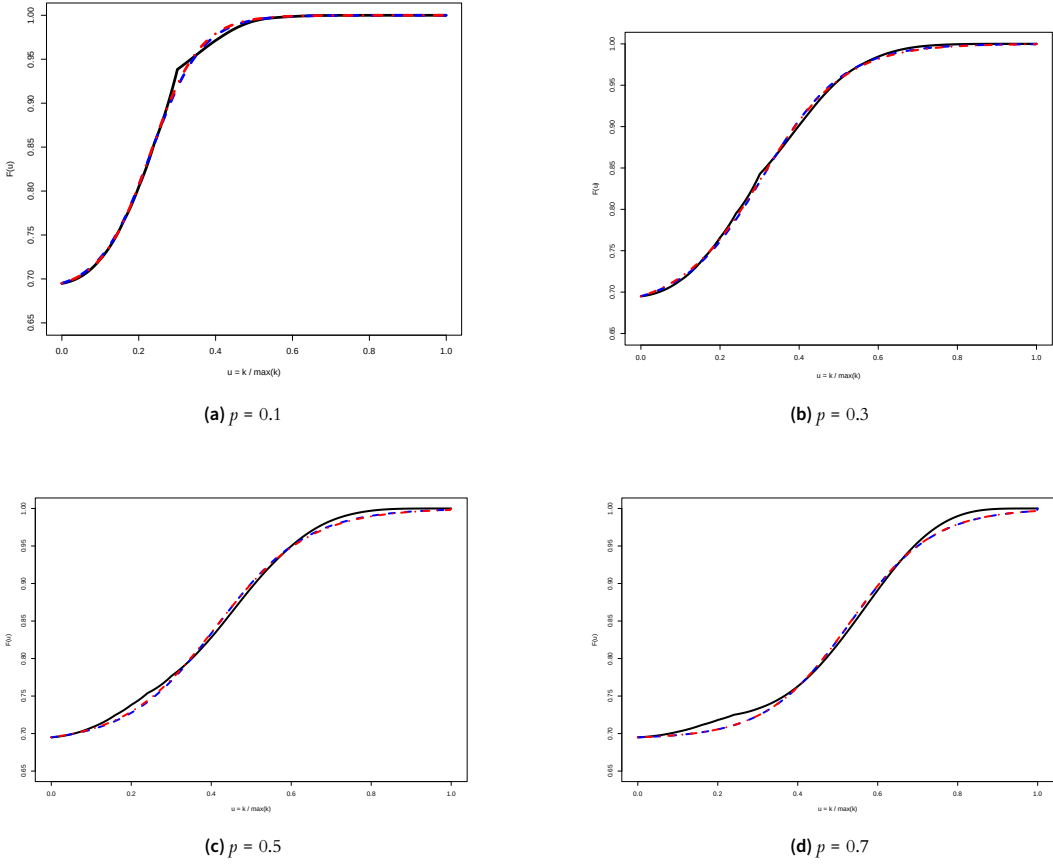


Figure 8. MBBEFD model fitting to empirical distribution functions for various values of p . Empirical cdf (black); MLE (blue); LSE (red).

In Figure 8 we plotted the fit of the MBBEFD model to the empirical cdfs for different values of the propagation probability p on tree $\mathcal{T}^{5,3}$. The empirical cdf is shown as a solid line, the maximum likelihood estimation (mle) fit appears as a dashed line (blue), and the least squares estimation (lse) fit is displayed as a dot dashed line (red). These plots illustrate how the quality of the model fit evolves as the value of p increases from 0.1 to 0.7. We can see that across the four values of p , the MBBEFD model provides a reasonable fit to the empirical cdfs in general. For low values of p (e.g., 0.1), the fit is accurate at the boundaries but exhibits minor deviations in the middle. We can also observe that as p increases (e.g., from 0.3 to 0.7), the fit quality slightly deteriorates, with both estimation methods exhibiting alternating patterns of overestimation and underestimation. These deviations become more pronounced for higher values of p , although the overall fitting behavior remains consistent between the mle and lse approaches.

6.6 Relevance for our work

Exposure rating is a fundamental tool in property and reinsurance, as it enables the quantification of risk according to the actual exposure to losses. In the literature, we have Riegel 2010 who demonstrates that using detailed risk profiles, such as location level profiles, improves pricing accuracy and ensures fairer assessment of fire risk. On his side, Desmedt et al. 2012 emphasizes that exposure-based approaches, particularly in excess of loss property reinsurance, provide a systematic framework to

account for heterogeneous exposures and the dependence structure among risks. More recently, Gatti and Wüthrich 2025 highlight the flexibility of the MBBEFD family to model truncated and censored insurance losses, illustrating its robustness for real-world applications. In our work, the exact cdfs of losses generated by the AFSM was derived. We also attempt to fit these exact curves with MBBEFD models and the MBBEFD was observed to fit this cdfs remarkably well. This demonstrates that our model provides a powerful and flexible tool for capturing both the propagation dynamics of fire losses and the resulting loss distributions.

7. Conclusion

In this paper, we have introduced a novel actuarial fire-spreading model based on tree-structured MRFs. We observed that our new model could provide a flexible and analytically tractable framework for assessing fire risk in interconnected structures. By modeling fire propagation probabilistically along the edges of a tree and conditioning on the fire ignition location, we derive explicit expressions for the distribution of total losses. After that, using this approach enabled us to analyze the impact of propagation probabilities on expected losses and risk measures, such as Value-at-Risk (VaR) and stop-loss premiums. We conclude from our results that the site's topology and the underlying propagation probabilities have a significant influence on the distribution of fire losses (as the propagation probability increases, the fire losses increase). In particular, we have noticed that more hub-like or centralized structures tend to exhibit higher contagion potential, amplifying the financial impact of a fire event. Using our model, we also conducted pairwise comparisons of structural configurations based on stochastic orders, providing a powerful tool for fire risk assessment, pricing, and mitigation strategies for these configurations. If we were to compare our approach to existing simulation-based approaches, our analytical framework enhances transparency, computational efficiency, and precision. It also reinforces the links between the research work of actuaries and civil engineers.

In future work, we aim to adapt and apply our ASFM in the construction insurance sector, particularly for innovative construction materials such as mass timber. According to Parodi 2023, construction insurance presents unique actuarial challenges due to the one-off nature of each project. Since there is no renewal history and limited individual data, traditional methods such as experience rating or rate change analysis are often inapplicable. Pricing must instead rely on project-specific features such as design, materials, duration, and contractor profile. Parodi 2023 also highlights the fact that, in construction insurance, policies typically cover non-annual periods spanning the whole construction lifecycle, with risks varying across stages from low exposure at inception (e.g., an empty lot) to increasing risk as structures take shape. This results in a non-uniform premium earning pattern, further complicated by a post-completion maintenance phase where residual risks may persist.

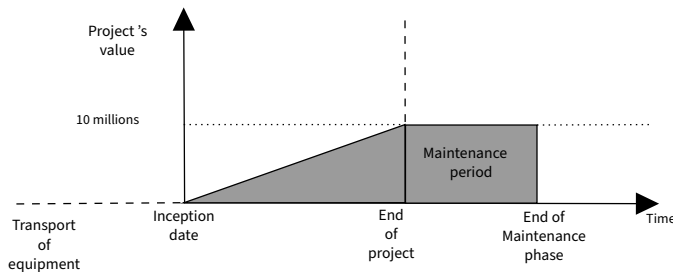


Figure 9. Coverage timeline of a construction insurance policy. Adapted from Figure 24.25 of Parodi 2023.

We adapted Figure 9 from Figure 24.25 of Parodi 2023 to illustrate this evolving exposure. We can notice that as the project advances, the value at risk grows, requiring pricing models that are

able to capture time-varying risk and irregular exposure. In short, effective pricing must shift focus from historical claims to present project characteristics and their temporal dynamics.

One can observe that the construction industry has embraced innovative materials to meet growing demands for sustainability, energy efficiency, and aesthetics over the past decade. Among them, mass timber has gained attention for its ecological and structural advantages. However, as stated before, its adoption poses new challenges for insurance ratemaking, particularly due to the lack of historical claims data, which leads to conservative pricing practices that may impede development. A pivotal moment occurred in 2021 when tall mass timber buildings were officially included in the International Building Code (IBC), permitting timber towers up to 18 stories in Canada and the U.S. This regulatory change has accelerated interest in mass timber, supported by initiatives like the Mass Timber Insurance Action Plan, led by the Canadian Wood Council and the Canadian Sustainable Buildings Alliance (CSBA and CWC).

Verisk, a significant data analytics and risk assessment company, categorizes its buildings into six categories (Frame to Fire Resistive) based on fire resistance and structural properties. So, traditional actuarial pricing relies on its construction classes. Yet, this system has faced criticism for its lack of methodological transparency, outdated assumptions, and misalignment with building codes in North America. In response, Verisk has announced a forthcoming class dedicated to mass timber, expected to be held in 2025–2026, see Shi and Kunzt 2025.

All this context highlights the relevance of our proposed AFSM, which addresses data scarcity. It takes into account fire test results from engineering studies such as Zhang, Mehaffey, and Hadjisophocleous 2015, Brandon and Östman 2016, Su et al. 2018, Brandon 2018; Brandon and Dagenais 2018, and Su et al. 2023. By doing so, our model integrates recent innovations in construction and fosters collaboration between actuaries and engineers in property and casualty (P&C) insurance.

The insurance industry is beginning to adapt. For example, Zurich Insurance now offers up to 50 million in coverage for mass timber projects (Zurich 2024), (Zurich 2021). Verisk also acknowledges this transformation in their publication "Mass Timber: New Technology Drives a New Construction Class", alongside increasing recognition of the material's potential through projects featured in "Top 25 Tallest Mass Timber Buildings in the World".

8. Acknowledgements

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Appendix 1. Tree structures representing a 4-unit building

In this appendix, we provide a detailed illustration of the different tree structures considered for a building with four units.

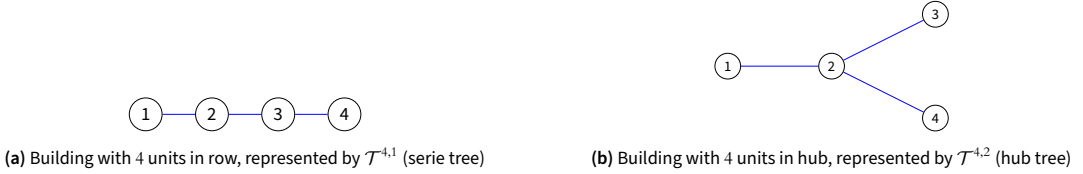


Figure 10. Different tree structures representing a 4-unit building

Appendix 2. Detailed analysis for the case $d=4$ units

The two possible tree configurations, denoted $\mathcal{T}^{4,1}$ and $\mathcal{T}^{4,2}$, are shown in Figure 10. The conditional stochastic representations of total losses given a fire ignition location to be s for each tree configuration $\mathcal{T}^{4,1}$ and $\mathcal{T}^{4,2}$, are given by

$$(B^{\mathcal{T}^{4,1}}|\Theta = s) = \begin{cases} (I_1|\Theta = 1) \left(Y_1 + (I_2|I_1 = 1)Y_2 + (I_2|I_1 = 1)(I_3|I_2 = 1)Y_3 + (I_2|I_1 = 1)(I_3|I_2 = 1)(I_4|I_3 = 1)Y_4 \right), & s = 1, \\ (I_2|\Theta = 2) \left(Y_2 + (I_1|I_2 = 1)Y_1 + (I_3|I_2 = 1)Y_3 + (I_3|I_2 = 1)(I_4|I_3 = 1)Y_4 \right), & s = 2, \\ (I_3|\Theta = 3) \left(Y_3 + (I_2|I_3 = 1)Y_2 + (I_4|I_3 = 1)Y_4 + (I_1|I_2 = 1)(I_2|I_3 = 1)Y_1 \right), & s = 3, \\ (I_4|\Theta = 4) \left(Y_4 + (I_3|I_4 = 1)Y_3 + (I_2|I_3 = 1)(I_3|I_4 = 1)Y_2 + (I_3|I_4 = 1)(I_2|I_3 = 1)(I_1|I_2 = 1)Y_1 \right), & s = 4, \end{cases}$$

and

$$(B^{\mathcal{T}^{4,2}}|\Theta = s) = \begin{cases} (I_1|\Theta = 1) \left(Y_1 + (I_2|I_1 = 1)Y_2 + (I_2|I_1 = 1)(I_3|I_2 = 1)Y_3 + (I_2|I_1 = 1)(I_4|I_2 = 1)Y_4 \right), & s = 1, \\ (I_2|\Theta = 2) \left(Y_2 + (I_1|I_2 = 1)Y_1 + (I_3|I_2 = 1)Y_3 + (I_4|I_2 = 1)Y_4 \right), & s = 2, \\ (I_3|\Theta = 3) \left(Y_3 + (I_2|I_3 = 1)Y_2 + (I_2|I_3 = 1)(I_1|I_2 = 1)Y_1 + (I_2|I_3 = 1)(I_4|I_2 = 1)Y_4 \right), & s = 3, \\ (I_4|\Theta = 4) \left(Y_4 + (I_2|I_4 = 1)Y_2 + (I_2|I_4 = 1)(I_1|I_2 = 1)Y_1 + (I_2|I_4 = 1)(I_3|I_2 = 1)Y_3 \right), & s = 4. \end{cases}$$

The cdfs of $(B^{\mathcal{T}^{4,1}}|\Theta = s)$ and $(B^{\mathcal{T}^{4,2}}|\Theta = s)$ are respectively given by

$$F_{B^{\mathcal{T}^{4,1}}|\Theta=s}(x) = \begin{cases} p_1(q_{1,2}F_{Y_1}(x) + p_{1,2}q_{2,3}F_{Y_{1+2}}(x) + p_{1,2}p_{2,3}q_{3,4}F_{Y_{1+2+3}}(x) + p_{1,2}p_{2,3}p_{3,4}F_{Y_{1+2+3+4}}(x)) + q_1, & s = 1, \\ p_2(q_{2,1}q_{2,3}F_{Y_2}(x) + p_{2,1}q_{2,3}F_{Y_{1+2}}(x) + q_{2,1}p_{2,3}q_{3,4}F_{Y_{2+3}}(x) + p_{2,1}p_{2,3}q_{3,4}F_{Y_{1+2+3}}(x) + q_{2,1}p_{2,3}p_{3,4}F_{Y_{2+3+4}}(x) \\ + p_{2,1}p_{2,3}p_{3,4}F_{Y_{1+2+3+4}}(x)) + q_2, & s = 2, \\ p_3(q_{3,2}q_{3,4}F_{Y_3}(x) + p_{3,2}q_{2,1}q_{3,4}F_{Y_{2+3}}(x) + q_{3,2}p_{3,4}F_{Y_{3+4}}(x) + p_{3,2}p_{2,1}q_{3,4}F_{Y_{1+2+3}}(x) + p_{3,2}q_{2,1}p_{3,4}F_{Y_{2+3+4}}(x) \\ + p_{3,2}p_{2,1}p_{3,4}F_{Y_{1+2+3+4}}(x)) + q_3, & s = 3, \\ p_4(q_{4,3}F_{Y_4}(x) + p_{4,3}q_{3,2}F_{Y_{3+4}}(x) + p_{4,3}p_{3,2}q_{2,1}F_{Y_{2+3+4}}(x) + p_{4,3}p_{3,2}p_{2,1}F_{Y_{1+2+3+4}}(x)) + q_4, & s = 4, \end{cases}$$

and

$$F_{B^{\mathcal{T}^{4,2}}|\Theta=s}(x) = \begin{cases} p_1(q_{12}F_{Y_1}(x) + p_{1,2}q_{2,3}q_{2,4}F_{Y_{1+2}}(x) + p_{1,2}q_{2,3}p_{2,4}F_{Y_{1+2+3}}(x) + p_{1,2}p_{2,3}q_{2,4}F_{Y_{1+2+3+4}}(x)) + q_1, & s = 1, \\ p_2(p_{2,1}q_{2,3}q_{2,4}F_{Y_{1+2}}(x) + p_{2,1}q_{2,3}p_{2,4}F_{Y_{1+2+3}}(x) + p_{2,1}p_{2,3}q_{2,4}F_{Y_{1+2+3+4}}(x) + p_{2,1}p_{2,3}p_{2,4}F_{Y_{1+2+3+4}}(x) \\ + q_{2,1}q_{2,3}q_{2,4}F_{Y_2}(x) + q_{2,1}q_{2,3}p_{2,4}F_{Y_{2+4}}(x) + q_{2,1}p_{2,3}q_{2,4}F_{Y_{2+3}}(x) + q_{2,1}p_{2,3}p_{2,4}F_{Y_{2+3+4}}(x)) + q_2, & s = 2, \\ p_3(p_{3,2}q_{2,4}p_{2,1}F_{Y_{1+2+3}}(x) + p_{3,2}p_{2,1}p_{2,4}F_{Y_{1+2+3+4}}(x) + q_{3,2}F_{Y_3}(x) + p_{3,2}q_{2,1}q_{2,4}F_{Y_{2+3}}(x) + p_{3,2}q_{2,1}p_{2,4}F_{Y_{2+3+4}}(x)) + q_3, & s = 3, \\ p_4(p_{4,2}p_{2,1}q_{2,3}F_{Y_{1+2+4}}(x) + p_{4,2}p_{2,1}p_{2,3}F_{Y_{1+2+3+4}}(x) + q_{4,2}F_{Y_4}(x) + p_{4,2}q_{2,1}q_{2,3}F_{Y_{2+4}}(x) + p_{4,2}q_{2,1}p_{2,3}F_{Y_{2+3+4}}(x)) + q_4, & s = 4. \end{cases}$$

At unit $v \in \mathcal{V}$, the rv I_v follows a Bernoulli distribution with parameter p_v^{bern} . That parameter admits the same decomposition as the one in (5.3) given by $p_v^{bern} = f_{\Theta}(v)p_v + \gamma_v$, $v \in \mathcal{V}$, where the

value γ_ν is given by

$$\gamma_\nu^{\mathcal{T}^{4,1}} = \begin{cases} f_{\Theta}^{\mathcal{T}^{4,1}}(2)p_2p_{2,1} + f_{\Theta}^{\mathcal{T}^{4,1}}(3)p_3p_{3,2}p_{2,1} + f_{\Theta}^{\mathcal{T}^{4,1}}(4)p_4p_{4,3}p_{3,2}p_{2,1}, & \nu = 1, \\ f_{\Theta}^{\mathcal{T}^{4,1}}(1)p_1p_{1,2} + f_{\Theta}^{\mathcal{T}^{4,1}}(3)p_3p_{3,2} + f_{\Theta}^{\mathcal{T}^{4,1}}(4)p_4p_{4,3}p_{3,2}, & \nu = 2, \\ f_{\Theta}^{\mathcal{T}^{4,1}}(1)p_1p_{1,2}p_{2,3} + f_{\Theta}^{\mathcal{T}^{4,1}}(2)p_2p_{2,3} + f_{\Theta}^{\mathcal{T}^{4,1}}(4)p_4p_{4,3}, & \nu = 3, \\ f_{\Theta}^{\mathcal{T}^{4,1}}(1)p_1p_{1,2}p_{2,3}p_{3,4} + f_{\Theta}^{\mathcal{T}^{4,1}}(2)p_2p_{2,3}p_{3,4} + f_{\Theta}^{\mathcal{T}^{4,1}}(3)p_3p_{3,4}, & \nu = 4, \end{cases}$$

and

$$\gamma_\nu^{\mathcal{T}^{4,2}} = \begin{cases} f_{\Theta}^{\mathcal{T}^{4,2}}(2)p_2p_{2,1} + f_{\Theta}^{\mathcal{T}^{4,2}}(3)p_3p_{3,2}p_{2,1} + f_{\Theta}^{\mathcal{T}^{4,2}}(4)p_4p_{4,2}p_{2,1}, & \nu = 1, \\ f_{\Theta}^{\mathcal{T}^{4,2}}(1)p_1p_{1,2} + f_{\Theta}^{\mathcal{T}^{4,2}}(3)p_3p_{3,2} + f_{\Theta}^{\mathcal{T}^{4,2}}(4)p_4p_{4,2}, & \nu = 2, \\ f_{\Theta}^{\mathcal{T}^{4,2}}(1)p_1p_{1,2}p_{2,3} + f_{\Theta}^{\mathcal{T}^{4,2}}(2)p_2p_{2,3} + f_{\Theta}^{\mathcal{T}^{4,2}}(4)p_4p_{4,2}p_{2,3}, & \nu = 3, \\ f_{\Theta}^{\mathcal{T}^{4,2}}(1)p_1p_{1,2}p_{2,4} + f_{\Theta}^{\mathcal{T}^{4,2}}(2)p_2p_{2,4} + f_{\Theta}^{\mathcal{T}^{4,2}}(3)p_3p_{3,2}p_{2,4}, & \nu = 4. \end{cases}$$

Under the assumptions A1-A4 of Section 5.4 in the context of $d = 4$, the cdfs of $B^{\mathcal{T}^{4,1}}$ and $B^{\mathcal{T}^{4,2}}$ are given by

$$F_{B^{\mathcal{T}^{4,1}}}(x) = \frac{q + q^2}{2}F_Y(x) + \left(pq + \frac{pq^2}{2}\right)F_Y^{*2}(x) + \frac{3p^2q}{2}F_Y^{*3}(x) + p^3F_Y^{*4}(x), \quad x \in [0, b],$$

and

$$F_{B^{\mathcal{T}^{4,2}}}(x) = \frac{3q + q^3}{4}F_Y(x) + \frac{3pq^2}{2}F_Y^{*2}(x) + \frac{9p^2q}{4}F_Y^{*3}(x) + p^3F_Y^{*4}(x), \quad x \in [0, b].$$

Also, the cdfs of $M^{\mathcal{T}^{4,1}}$ and $M^{\mathcal{T}^{4,2}}$ are given by

$$F_{M^{\mathcal{T}^{4,1}}}(x) = \begin{cases} 1 - \frac{3}{2}p + \frac{1}{2}p^2 & 1 \leq x < 2 \\ 1 - \frac{3}{2}p^2 + \frac{1}{2}p^3 & 2 \leq x < 3, \\ 1 - p^3 & 3 \leq x < 4 \end{cases} \quad \text{and} \quad F_{M^{\mathcal{T}^{4,2}}}(x) = \begin{cases} \frac{1}{4}(3 - 3p + (1-p)^3) & 1 \leq x < 2 \\ 1 - \frac{1}{4}(9p^2 - 5p^3) & 2 \leq x < 3, \\ 1 - p^3 & 3 \leq x < 4 \end{cases}$$

with $F_{M^{\mathcal{T}^{4,1}}}(x) = F_{M^{\mathcal{T}^{4,2}}}(x) = 0$ for $0 \leq x < 1$ and $= 1$ for $x \geq 4$.

Appendix 3. Summary Tables of risk measures and conditional distributions

Table 5. $f_{I|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{4,1}$

i_1	i_2	i_3	i_4	$f_{I \Theta=1}(i)$	$f_{I \Theta=2}(i)$	$f_{I \Theta=3}(i)$	$f_{I \Theta=4}(i)$
1	0	0	0	$p_1 q_{1 \rightarrow 2}$	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	$p_1 p_{1 \rightarrow 2} q_{2 \rightarrow 3}$	$p_2 p_{2 \rightarrow 1} q_{2 \rightarrow 3}$	0	0
1	1	0	1	0	0	0	0
1	1	1	0	$p_1 p_{1 \rightarrow 2} p_{2 \rightarrow 3} q_{3 \rightarrow 4}$	$p_2 p_{2 \rightarrow 3} p_{2 \rightarrow 1} q_{3 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} p_{2 \rightarrow 1} q_{3 \rightarrow 4}$	0
1	1	1	1	$p_1 p_{1 \rightarrow 2} p_{2 \rightarrow 3} p_{3 \rightarrow 4}$	$p_2 p_{2 \rightarrow 3} p_{2 \rightarrow 1} p_{3 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} p_{2 \rightarrow 1} p_{3 \rightarrow 4}$	$p_4 p_{4 \rightarrow 3} p_{3 \rightarrow 2} p_{2 \rightarrow 1}$
0	0	0	0	q_1	q_2	q_3	q_4
0	0	0	1	0	0	0	$p_4 q_{4 \rightarrow 3}$
0	0	1	0	0	0	$p_3 q_{3 \rightarrow 2} q_{3 \rightarrow 4}$	0
0	0	1	1	0	0	$p_3 q_{3 \rightarrow 2} p_{3 \rightarrow 4}$	$p_4 p_{4 \rightarrow 3} q_{3 \rightarrow 2}$
0	1	0	0	0	$p_2 q_{2 \rightarrow 3} q_{2 \rightarrow 1}$	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	$p_2 p_{2 \rightarrow 3} q_{2 \rightarrow 1} q_{3 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} q_{2 \rightarrow 1} q_{3 \rightarrow 4}$	0
0	1	1	1	0	$p_2 p_{2 \rightarrow 3} q_{2 \rightarrow 1} p_{3 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} q_{2 \rightarrow 1} p_{3 \rightarrow 4}$	$p_4 p_{4 \rightarrow 3} p_{3 \rightarrow 2} q_{2 \rightarrow 1}$

Table 6. $f_{I|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{4,2}$

i_1	i_2	i_3	i_4	$f_{I \Theta=1}(i)$	$f_{I \Theta=2}(i)$	$f_{I \Theta=3}(i)$	$f_{I \Theta=4}(i)$
1	0	0	0	$p_1 q_{1 \rightarrow 2}$	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	$p_1 p_{1 \rightarrow 2} q_{2 \rightarrow 3} q_{2 \rightarrow 4}$	$p_2 p_{2 \rightarrow 1} q_{2 \rightarrow 3} q_{2 \rightarrow 4}$	0	0
1	1	0	1	$p_1 p_{1 \rightarrow 2} q_{2 \rightarrow 3} p_{2 \rightarrow 4}$	$p_2 p_{2 \rightarrow 1} q_{2 \rightarrow 3} p_{2 \rightarrow 4}$	0	$p_4 p_{4 \rightarrow 2} p_{2 \rightarrow 1} q_{2 \rightarrow 3}$
1	1	1	0	$p_1 p_{1 \rightarrow 2} p_{2 \rightarrow 3} q_{2 \rightarrow 4}$	$p_2 p_{2 \rightarrow 1} p_{2 \rightarrow 3} q_{2 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} p_{2 \rightarrow 1} q_{2 \rightarrow 4}$	0
1	1	1	1	$p_1 p_{1 \rightarrow 2} p_{2 \rightarrow 3} p_{2 \rightarrow 4}$	$p_2 p_{2 \rightarrow 1} p_{2 \rightarrow 3} p_{2 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} p_{2 \rightarrow 1} p_{2 \rightarrow 4}$	$p_4 p_{4 \rightarrow 2} p_{2 \rightarrow 1} p_{2 \rightarrow 3}$
0	0	0	0	q_1	q_2	q_3	q_4
0	0	0	1	0	0	0	$p_4 q_{4 \rightarrow 2}$
0	0	1	0	0	0	$p_3 q_{3 \rightarrow 2}$	0
0	0	1	1	0	0	0	0
0	1	0	0	0	$p_2 q_{2 \rightarrow 1} q_{2 \rightarrow 3} q_{2 \rightarrow 4}$	0	0
0	1	0	1	0	$p_2 q_{2 \rightarrow 1} q_{2 \rightarrow 3} p_{2 \rightarrow 4}$	0	$p_4 p_{4 \rightarrow 2} q_{2 \rightarrow 1} q_{2 \rightarrow 3}$
0	1	1	0	0	$p_2 q_{2 \rightarrow 1} p_{2 \rightarrow 3} q_{2 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} q_{2 \rightarrow 1} q_{2 \rightarrow 4}$	0
0	1	1	1	0	$p_2 q_{2 \rightarrow 1} p_{2 \rightarrow 3} p_{2 \rightarrow 4}$	$p_3 p_{3 \rightarrow 2} q_{2 \rightarrow 1} p_{2 \rightarrow 4}$	$p_4 p_{4 \rightarrow 2} q_{2 \rightarrow 1} p_{2 \rightarrow 3}$

Appendix 4. Conditional joint pmf of the Bernoulli random vectors for d=5**Table 7.** $f_{\Pi|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,1}$

$i_1 i_2 i_3 i_4 i_5$	$f_{\Pi \Theta=1}(i)$	$f_{\Pi \Theta=2}(i)$	$f_{\Pi \Theta=3}(i)$	$f_{\Pi \Theta=4}(i)$	$f_{\Pi \Theta=5}(i)$
10000	$p_1 q_{1+2}$	0	0	0	0
10001	0	0	0	0	0
10010	0	0	0	0	0
10011	0	0	0	0	0
10100	0	0	0	0	0
10101	0	0	0	0	0
10110	0	0	0	0	0
10111	0	0	0	0	0
11000	$p_1 p_{1+2} q_{2+3}$	$p_2 p_{2+1} q_{2+3}$	0	0	0
11001	0	0	0	0	0
11010	0	0	0	0	0
11011	0	0	0	0	0
11100	$p_1 p_{1+2} p_{2+3} q_{3+4}$	$p_2 p_{2+1} p_{2+3} q_{3+4}$	$p_3 p_{3+2} p_{2+1} q_{3+4}$	0	0
11101	0	0	0	0	0
11110	$p_1 p_{1+2} p_{2+3} p_{3+4} q_{4+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} q_{4+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} q_{4+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} q_{4+5}$	0
11111	$p_1 p_{1+2} p_{2+3} p_{3+4} p_{4+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} p_{4+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} p_{4+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} p_{4+5}$	$p_5 p_{5+4} p_{4+3} p_{3+2} p_{2+1}$
00000	q_1	q_2	q_3	q_4	q_5
00001	0	0	0	0	$p_5 q_{5+4}$
00010	0	0	0	$p_4 q_{4+3} q_{4+5}$	0
00011	0	0	0	$p_4 q_{4+3} p_{4+5}$	$p_5 p_{5+4} q_{4+3}$
00100	0	0	$p_3 q_{3+2} q_{3+4}$	0	0
00101	0	0	0	0	0
00110	0	0	$p_3 q_{3+2} p_{3+4} q_{4+5}$	$p_4 p_{4+3} q_{3+2} q_{4+5}$	0
00111	0	0	$p_3 q_{3+2} p_{3+4} p_{4+5}$	$p_4 p_{4+3} q_{3+2} p_{4+5}$	$p_5 p_{5+4} p_{4+3} q_{3+2}$
01000	0	$p_2 q_{2+1} q_{2+3}$	0	0	0
01001	0	0	0	0	0
01010	0	0	0	0	0
01011	0	0	0	0	0
01100	0	$p_2 q_{2+1} p_{2+3} q_{3+4}$	$p_3 p_{3+2} q_{2+1} q_{3+4}$	0	0
01101	0	0	0	0	0
01110	0	$p_2 q_{2+1} p_{2+3} p_{3+4} q_{4+5}$	$p_3 p_{3+2} q_{2+1} p_{3+4} q_{4+5}$	$p_4 p_{4+3} p_{3+2} q_{2+1} q_{4+5}$	0
01111	0	$p_2 p_{2+1} p_{2+3} p_{3+4} p_{4+5}$	$p_3 p_{3+2} q_{2+1} p_{3+4} p_{4+5}$	$p_4 p_{4+3} p_{3+2} q_{2+1} p_{4+5}$	$p_5 p_{5+4} p_{4+3} p_{3+2} p_{2+1}$

Table 8. $f_{\Pi|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,1}$ assuming that $\Pr(Y_s = 1) = 1$, for all $s \in \mathcal{V}$

$f_{\Pi \Theta=s}(k)$	$s = 1$	$s = 2$	$s = 3$	$s = 4$	$s = 5$
$k = 0$	q_1	q_2	q_3	q_4	q_5
$k = 1$	$p_1 q_{1+2}$	$p_2 q_{2+1} q_{2+3}$	$p_3 q_{3+2} q_{3+4}$	$p_4 q_{4+3} q_{4+5}$	$p_5 q_{5+4}$
$k = 2$	$p_1 p_{1+2} q_{2+3}$	$p_2 (p_{2+1} q_{2+3} + q_{2+1} p_{2+3} q_{3+4})$	$p_3 (p_{3+2} q_{2+1} q_{3+4} + q_{3+2} p_{3+4} q_{4+5})$	$p_4 (p_{4+3} q_{3+2} q_{4+5} + p_{4+3} q_{3+2} p_{4+5})$	$p_5 p_{5+4} q_{4+3}$
$k = 3$	$p_1 p_{1+2} p_{2+3} q_{3+4}$	$p_2 (p_{2+1} p_{2+3} q_{3+4} + q_{2+1} p_{2+3} p_{3+4} q_{4+5})$	$p_3 (q_{3+2} p_{3+4} p_{4+5} + p_{3+2} p_{2+1} q_{3+4} + p_{3+2} q_{2+1} p_{3+4} q_{4+5})$	$p_4 (p_{4+3} p_{4+5} q_{3+2} + p_{4+3} p_{3+2} q_{2+1} p_{4+5})$	$p_5 p_{5+4} p_{4+3} q_{3+2}$
$k = 4$	$p_1 p_{1+2} p_{2+3} p_{3+4} q_{4+5}$	$p_2 (p_{2+1} p_{2+3} p_{3+4} q_{4+5} + q_{2+1} p_{2+3} p_{3+4} p_{4+5})$	$p_3 (p_{3+2} q_{2+1} p_{3+4} p_{4+5} + p_{3+2} p_{2+1} p_{3+4} q_{4+5})$	$p_4 (p_{4+3} p_{3+2} p_{2+1} q_{4+5} + p_{4+3} p_{3+2} q_{2+1} p_{4+5})$	$p_5 p_{5+4} p_{4+3} p_{3+2} q_{2+1}$
$k = 5$	$p_1 p_{1+2} p_{2+3} p_{3+4} p_{4+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} p_{4+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} p_{4+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} p_{4+5}$	$p_5 p_{5+4} p_{4+3} p_{3+2} p_{2+1}$

Table 9. $f_{I|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,2}$

$i_1 i_2 i_3 i_4 i_5$	$f_{I \Theta=1}(i)$	$f_{I \Theta=2}(i)$	$f_{I \Theta=3}(i)$	$f_{I \Theta=4}(i)$	$f_{I \Theta=5}(i)$
10000	$p_1 q_{1+2}$	0	0	0	0
10001	0	0	0	0	0
10010	0	0	0	0	0
10011	0	0	0	0	0
10100	0	0	0	0	0
10101	0	0	0	0	0
10110	0	0	0	0	0
10111	0	0	0	0	0
11000	$p_1 p_{1+2} q_{2+3}$	$p_2 p_{2+1} q_{2+3}$	0	0	0
11001	0	0	0	0	0
11010	0	0	0	0	0
11011	0	0	0	0	0
11100	$p_1 p_{1+2} p_{2+3} q_{3+4} q_{3+5}$	$p_2 p_{2+1} p_{2+3} q_{3+4} q_{3+5}$	$p_3 p_{3+2} p_{2+1} q_{3+4} q_{3+5}$	0	0
11101	$p_1 p_{1+2} p_{2+3} q_{3+4} p_{3+5}$	$p_2 p_{2+1} p_{2+3} q_{3+4} p_{3+5}$	$p_3 p_{3+2} p_{2+1} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} p_{3+2} p_{2+1} q_{3+4}$
11110	$p_1 p_{1+2} p_{2+3} p_{3+4} q_{3+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} q_{3+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} q_{3+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} q_{3+5}$	0
11111	$p_1 p_{1+2} p_{2+3} p_{3+4} p_{3+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} p_{3+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} p_{3+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} p_{3+5}$	$p_5 p_{5+3} p_{3+2} p_{2+1} p_{3+4}$
00000	q_1	q_2	q_3	q_4	q_5
00001	0	0	0	0	$p_5 q_{5+3}$
00010	0	0	0	$p_4 q_{4+3}$	0
00011	0	0	0	0	0
00100	0	0	$p_3 q_{3+2} q_{3+4} q_{3+5}$	0	0
00101	0	0	$p_3 q_{3+2} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} q_{3+2} q_{3+4}$
00110	0	0	$p_3 q_{3+2} p_{3+4} q_{3+5}$	$p_4 p_{4+3} q_{3+2} q_{3+5}$	0
00111	0	0	$p_3 q_{3+2} p_{3+4} p_{3+5}$	$p_4 p_{4+3} q_{3+2} p_{3+5}$	$p_5 p_{5+3} q_{3+2} p_{3+4}$
01000	0	$p_2 q_{2+1} q_{2+3}$	0	0	0
01001	0	0	0	0	0
01010	0	0	0	0	0
01011	0	0	0	0	0
01100	0	$p_2 q_{2+1} p_{2+3} q_{3+4} q_{3+5}$	$p_3 p_{3+2} q_{2+1} q_{3+4} q_{3+5}$	0	0
01101	0	$p_2 q_{2+1} p_{2+3} q_{3+4} p_{3+5}$	$p_3 p_{3+2} q_{2+1} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} p_{3+2} q_{2+1} q_{3+4}$
01110	0	$p_2 q_{2+1} p_{2+3} p_{3+4} q_{3+5}$	$p_3 p_{3+2} q_{2+1} p_{3+4} q_{3+5}$	$p_4 p_{4+3} p_{3+2} q_{2+1} q_{3+5}$	0
01111	0	$p_2 q_{2+1} p_{2+3} p_{3+4} p_{3+5}$	$p_3 p_{3+2} q_{2+1} p_{3+4} p_{3+5}$	$p_4 p_{4+3} p_{3+2} q_{2+1} p_{3+5}$	$p_5 p_{5+3} p_{3+2} q_{2+1} p_{3+4}$

Table 10. $f_{B|\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,2}$ assuming that $\Pr(Y_s = 1) = 1$, for all $s \in \mathcal{V}$

$f_{B \Theta=s}(k)$	$s = 1$	$s = 2$	$s = 3$	$s = 4$	$s = 5$
$k = 0$	q_1	q_2	q_3	q_4	q_5
$k = 1$	$p_1 q_{1+2}$	$p_2 q_{2+1} q_{2+3}$	$p_3 q_{3+2} q_{3+4} q_{3+5}$	$p_4 q_{4+3}$	$p_5 q_{5+3}$
$k = 2$	$p_1 p_{1+2} q_{2+3}$	$p_2 \left(p_{2+1} q_{2+3} + q_{2+1} p_{2+3} q_{3+4} q_{3+5} \right)$	$p_3 \left(p_{3+2} q_{2+1} q_{3+4} q_{3+5} + q_{3+2} p_{3+4} q_{3+5} + q_{3+2} q_{3+4} p_{3+5} \right)$	$p_4 p_{4+3} q_{3+2} q_{3+5}$	$p_5 p_{5+3} q_{3+2} q_{3+4}$
$k = 3$	$p_1 p_{1+2} p_{2+3} q_{3+4} q_{3+5}$	$p_2 \left(p_{2+1} p_{2+3} q_{3+4} q_{3+5} + q_{2+1} p_{2+3} q_{3+4} p_{3+5} + q_{2+1} p_{2+3} p_{3+4} q_{3+5} \right)$	$p_3 \left(q_{3+2} p_{3+4} p_{3+5} + p_{3+2} q_{2+1} q_{3+4} p_{3+5} + p_{3+2} q_{2+1} p_{3+4} q_{3+5} + p_{3+2} p_{2+1} q_{3+4} q_{3+5} \right)$	$p_4 \left(p_{4+3} q_{3+2} p_{3+5} + p_{4+3} q_{2+1} p_{3+2} q_{3+5} \right)$	$p_5 \left(p_{5+3} p_{3+4} q_{3+2} + p_{5+3} p_{3+2} q_{2+1} q_{3+4} \right)$
$k = 4$	$p_1 \left(p_{1+2} p_{2+3} p_{3+4} q_{3+5} + p_{1+2} p_{2+3} q_{3+4} p_{3+5} \right)$	$p_2 \left(p_{2+1} p_{2+3} p_{3+4} q_{3+5} + p_{2+1} p_{2+3} q_{3+4} p_{3+5} + q_{2+1} p_{2+3} p_{3+4} p_{3+5} \right)$	$p_3 \left(p_{3+2} p_{2+1} q_{3+4} p_{3+5} + p_{3+2} p_{2+1} p_{3+4} q_{3+5} + p_{3+2} q_{2+1} p_{3+4} p_{3+5} \right)$	$p_4 \left(p_{4+3} p_{3+2} q_{3+5} p_{2+1} + p_{4+3} q_{2+1} p_{3+2} p_{3+5} \right)$	$p_5 \left(p_{5+3} p_{2+1} p_{3+2} q_{3+4} + p_{5+3} p_{3+2} p_{2+1} q_{2+1} \right)$
$k = 5$	$p_1 p_{1+2} p_{2+3} p_{3+4} p_{3+5}$	$p_2 p_{2+1} p_{2+3} p_{3+4} p_{3+5}$	$p_3 p_{3+2} p_{2+1} p_{3+4} p_{3+5}$	$p_4 p_{4+3} p_{3+2} p_{2+1} p_{3+5}$	$p_5 p_{5+3} p_{2+1} p_{3+2} p_{3+4}$

Table 11. $f_{I\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,3}$

$i_1 i_2 i_3 i_4 i_5$	$f_{I\Theta=1}(i)$	$f_{I\Theta=2}(i)$	$f_{I\Theta=3}(i)$	$f_{I\Theta=4}(i)$	$f_{I\Theta=5}(i)$
10000	$p_1 q_{1+3}$	0	0	0	0
10001	0	0	0	0	0
10010	0	0	0	0	0
10011	0	0	0	0	0
10100	$p_1 q_{3+2} p_{1+3} q_{3+4} q_{3+5}$	0	$p_3 p_{3+1} q_{3+2} q_{3+4} q_{3+5}$	0	0
10101	$p_1 q_{3+2} p_{1+3} q_{3+4} p_{3+5}$	0	$p_3 p_{3+1} q_{3+2} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} q_{3+2} p_{3+1} q_{3+4}$
10110	$p_1 q_{3+2} p_{1+3} p_{3+4} q_{3+5}$	0	$p_3 p_{3+1} q_{3+2} p_{3+4} q_{3+5}$	$p_4 p_{3+1} q_{3+2} p_{4+3} q_{3+5}$	0
10111	$p_1 q_{3+2} p_{1+3} p_{3+4} p_{3+5}$	0	$p_3 p_{3+1} q_{3+2} p_{3+4} p_{3+5}$	$p_4 p_{3+1} q_{3+2} p_{4+3} p_{3+5}$	$p_5 p_{5+3} q_{3+2} p_{3+1} p_{3+4}$
11000	0	0	0	0	0
11001	0	0	0	0	0
11010	0	0	0	0	0
11011	0	0	0	0	0
11100	$p_1 p_{3+2} p_{1+3} q_{3+4} q_{3+5}$	$p_2 p_{2+3} p_{3+1} q_{3+4} q_{3+5}$	$p_3 p_{3+1} p_{3+2} q_{3+4} q_{3+5}$	0	0
11101	$p_1 p_{3+2} p_{1+3} q_{3+4} p_{3+5}$	$p_2 p_{2+3} p_{3+1} q_{3+4} p_{3+5}$	$p_3 p_{3+1} p_{3+2} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} p_{3+2} p_{3+1} q_{3+4}$
11110	$p_1 p_{3+2} p_{1+3} p_{3+4} q_{3+5}$	$p_2 p_{2+3} p_{3+1} p_{3+4} q_{3+5}$	$p_3 p_{3+1} p_{3+2} p_{3+4} q_{3+5}$	$p_4 p_{3+1} p_{3+2} p_{4+3} q_{3+5}$	0
11111	$p_1 p_{3+2} p_{1+3} p_{3+4} p_{3+5}$	$p_2 p_{2+3} p_{3+1} p_{3+4} p_{3+5}$	$p_3 p_{3+1} p_{3+2} p_{3+4} p_{3+5}$	$p_4 p_{3+1} p_{3+2} p_{4+3} p_{3+5}$	$p_5 p_{5+3} p_{3+2} p_{3+1} p_{3+4}$
00000	q_1	q_2	q_3	q_4	q_5
00001	0	0	0	0	$p_5 q_{5+3}$
00010	0	0	0	$p_4 q_{4+3}$	0
00011	0	0	0	0	0
00100	0	0	$p_3 q_{3+1} q_{3+2} q_{3+4} q_{3+5}$	0	0
00101	0	0	$p_3 q_{3+1} q_{3+2} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} q_{3+2} q_{3+1} q_{3+4}$
00110	0	0	$p_3 q_{3+1} q_{3+2} p_{3+4} q_{3+5}$	$p_4 q_{3+1} q_{3+2} p_{4+3} q_{3+5}$	0
00111	0	0	$p_3 q_{3+1} q_{3+2} p_{3+4} p_{3+5}$	$p_4 q_{3+1} q_{3+2} p_{4+3} p_{3+5}$	$p_5 p_{5+3} q_{3+2} q_{3+1} p_{3+4}$
01000	0	$p_2 q_{2+3}$	0	0	0
01001	0	0	0	0	0
01010	0	0	0	0	0
01011	0	0	0	0	0
01100	0	$p_2 p_{2+3} q_{3+1} q_{3+4} q_{3+5}$	$p_3 q_{3+1} p_{3+2} q_{3+4} q_{3+5}$	0	0
01101	0	$p_2 p_{2+3} q_{3+1} q_{3+4} p_{3+5}$	$p_3 q_{3+1} p_{3+2} q_{3+4} p_{3+5}$	0	$p_5 p_{5+3} p_{3+2} q_{3+1} q_{3+4}$
01110	0	$p_2 p_{2+3} q_{3+1} p_{3+4} q_{3+5}$	$p_3 q_{3+1} p_{3+2} p_{3+4} q_{3+5}$	$p_4 q_{3+1} p_{3+2} p_{4+3} q_{3+5}$	0
01111	0	$p_2 p_{2+3} q_{3+1} p_{3+4} p_{3+5}$	$p_3 q_{3+1} p_{3+2} p_{3+4} p_{3+5}$	$p_4 q_{3+1} p_{3+2} p_{4+3} p_{3+5}$	$p_5 p_{5+3} p_{3+2} p_{3+1} p_{3+4}$

Table 12. Values of $f_{B|I\Theta=s}$, for $s \in \mathcal{V}$, for the tree $\mathcal{T}^{5,3}$ assuming that $\Pr(Y_s = 1) = 1$, for all $s \in \mathcal{V}$

$f_{B I\Theta=s}(k)$	$s = 1$	$s = 2$	$s = 3$	$s = 4$	$s = 5$
$k = 0$	q_1	q_2	q_3	q_4	q_5
$k = 1$	$p_1 q_{1+3}$	$p_2 q_{2+3}$	$p_3 q_{3+1} q_{3+2} q_{3+4} q_{3+5}$	$p_4 q_{4+3}$	$p_5 q_{5+3}$
$k = 2$	$p_1 p_{1+3} q_{3+2} q_{3+4} q_{3+5}$	$p_2 p_{2+3} q_{3+1} q_{3+4} q_{3+5}$	$p_3 \left(p_{3+1} q_{3+2} q_{3+4} q_{3+5} + \right. \\ \left. q_{3+1} p_{3+2} q_{3+4} q_{3+5} + \right. \\ \left. q_{3+1} q_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. q_{3+1} q_{3+2} q_{3+4} p_{3+5} \right)$	$p_4 p_{4+3} q_{3+1} q_{3+2} q_{3+5}$	$p_5 p_{5+3} q_{3+1} q_{3+2} q_{3+4}$
$k = 3$	$p_1 \left(p_{1+3} p_{3+2} q_{3+4} q_{3+5} + \right. \\ \left. p_{1+3} q_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. p_{1+3} q_{3+2} q_{3+4} p_{3+5} \right)$	$p_2 \left(p_{2+3} p_{3+1} q_{3+4} q_{3+5} + \right. \\ \left. p_{2+3} q_{3+1} p_{3+4} q_{3+5} + \right. \\ \left. p_{2+3} q_{3+1} q_{3+4} p_{3+5} \right)$	$p_3 \left(p_{3+1} p_{3+2} q_{3+4} q_{3+5} + \right. \\ \left. p_{3+1} q_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. p_{3+1} q_{3+2} q_{3+4} p_{3+5} + \right. \\ \left. q_{3+1} p_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. q_{3+1} p_{3+2} q_{3+4} p_{3+5} \right)$	$p_4 \left(p_{4+3} p_{3+1} q_{3+2} q_{3+5} + \right. \\ \left. p_{4+3} q_{3+1} p_{3+2} q_{3+5} + \right. \\ \left. p_{4+3} q_{3+1} q_{3+2} p_{3+5} \right)$	$p_5 \left(p_{5+3} p_{3+1} q_{3+2} q_{3+4} + \right. \\ \left. p_{5+3} q_{3+1} p_{3+2} q_{3+4} + \right. \\ \left. p_{5+3} q_{3+1} q_{3+2} p_{3+4} \right)$
$k = 4$	$p_1 \left(p_{1+3} p_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. p_{1+3} p_{3+2} q_{3+4} p_{3+5} \right)$	$p_2 \left(p_{2+3} p_{3+1} p_{3+4} q_{3+5} + \right. \\ \left. p_{2+3} p_{3+1} q_{3+4} p_{3+5} \right)$	$p_3 \left(p_{3+1} p_{3+2} p_{3+4} q_{3+5} + \right. \\ \left. p_{3+1} q_{3+2} p_{3+4} p_{3+5} + \right. \\ \left. p_{3+1} p_{3+2} q_{3+4} p_{3+5} \right)$	$p_4 \left(p_{4+3} p_{3+1} p_{3+2} q_{3+5} + \right. \\ \left. p_{4+3} q_{3+1} p_{3+2} p_{3+5} \right)$	$p_5 \left(p_{5+3} p_{3+1} p_{3+2} q_{3+4} + \right. \\ \left. p_{5+3} q_{3+1} p_{3+2} p_{3+4} \right)$
$k = 5$	$p_1 p_{1+3} p_{3+2} p_{3+4} p_{3+5}$	$p_2 p_{2+3} p_{3+1} p_{3+4} p_{3+5}$	$p_3 p_{3+1} p_{3+2} p_{3+4} p_{3+5}$	$p_4 p_{4+3} p_{3+1} p_{3+2} p_{3+5}$	$p_5 p_{5+3} p_{3+1} p_{3+4} p_{3+5}$

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